Pointwise Complexity for Gaussian Fields: Upper Envelopes, Algorithmic Lower Bounds, and Finite-Cutoff Renormalization

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Abstract

We prove a variance-aware pointwise majorizing-measure theorem for centered Gaussian processes. Classical generic chaining characterizes the scalar quantity $\mathbb{E}\sup_{x\in T} X_x$; the theorem here gives a simultaneous high-probability envelope for the entire field. For an ambient prior μ, the envelope at x is governed by a localized Fernique–Talagrand functional $\Phi_\mu(x)$, truncated at the variance scale $\sigma(x) = (\mathbb{E}X_x^2)^{1/2}$, together with the corresponding Gaussian tail term. Optimizing the resulting in-expectation bound recovers the anchored Fernique–Talagrand scale up to universal constants. The theorem is intended as a standard reference whenever one needs a field-level refinement of classical generic chaining, and as the Gaussian-process counterpart of pointwise empirical-process bounds for deep neural networks (Li and Xu, 2026).

We also derive a single-radius Bayesian algorithmic lower envelope from the interactive Fano/data-processing principle (Chen et al., 2024). For a known prior, an observation channel, and a concrete estimator $\widehat{t}(Y)$, the lower bound is expressed through the exact ghost small-ball mass $\mathbb{E}_{Y\sim Q}\pi(B_d(\widehat{t}(Y),\Delta))$, rather than a worst-case covering number. In Gaussian location experiments, comparison decoders convert Bayes location error into lower bounds on decision-aligned Gaussian ranges. This includes ordinary nearest-neighbor decoding and, importantly in overparameterized ambient classes, unrestricted norm-regularized nearest-neighbor rules. The resulting lower bound is valid for arbitrary priors, with magnitude controlled by the ghost-mass-induced pointwise dimension. For hard priors localized at the tested radius, a class-supremum relaxation recovers a Sudakov-type single-radius obstruction. We then construct an elementary heterogeneous-orthogonal-basis example separating the usual Fano relaxation for a fixed prior, the Bayesian algorithmic lower envelope, the pointwise Gaussian envelope on the estimator-selected subatlas, and the full-class minimax risk/global Gaussian scale. Together, these results show that Bayesian algorithmic lower bounds can serve as local-geometric certificates of pointwise complexity for fixed estimators in overparameterized ambient classes.

We further isolate a pointwise-complexity principle for statistical field theory and finite-cutoff renormalization, guided by probabilistic and physical motivation. For renormalization maps and interacting fields, we prove finite-cutoff theorems and construct a nonconvex phase-atlas example, illustrating a route to controlling coarse marginals of globally non-log-concave and anisotropic measures beyond the scope of traditional convex geometry. An exact finite-step telescoping identity replaces one renormalization level at a time; the resulting secant response Gram defines a pointwise ellipsoidal metric; and a hierarchical prior over phase labels and active subspaces splits the complexity into a local effective dimension and a global atlas cost. This makes precise the structural analogy between exponential-family RG and nonlinear feature learning in fully connected DNNs (Li and Xu, 2026): both settings exploit a composite linear structure. For free fields and local time, we specialize the pointwise Gaussian theorem to finite pinned Gaussian free fields. The resulting target-set local-time envelopes explicitly separate direct field-level control from reductions through full cover time. On phase-star graphs, a selected subatlas pays its own atlas code length and chart cost, while any full-cover route must pay for all phases.

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1 Introduction and Main Results

Classical generic chaining controls the scalar quantity $\mathbb{E}\sup_{x\in T} X_x$ for a Gaussian process $\{X_x\}_{x\in T}$. For many field-level questions, however, this scalar is not the first object one wants to estimate. In the graph setting, the second Ray–Knight theorem identifies a local-time field at an inverse-local-time stopping rule with a shifted square of a Gaussian free field. In constructive and lattice field theory, one similarly studies local observables before passing to global maxima or continuum limits. This suggests that the Gaussian input should itself be pointwise: one should first prove a simultaneous field-level envelope and only afterwards take suprema or threshold events.

A further motivation comes from recent work on pointwise generalization for deep neural networks (Li and Xu, 2026). There the central object is not a class-wide supremum but a hypothesis-dependent finite-scale complexity. The proof is structural: an exact non-perturbative telescoping expansion exposes learned feature Gram matrices, and a hierarchical local-chart/global-atlas prior converts the resulting data-dependent active subspaces into a data-independent bound. The present paper develops the Gaussian-process counterpart of that viewpoint. Theorem 1.1 can serve as a standard reference whenever one needs a field-level refinement of classical generic chaining.

We also record a single-radius Bayesian lower-envelope counterpart. It is not a pathwise lower bound on each Gaussian coordinate. Rather, it is a data-processing Bayes-risk lower bound for a specified prior, observation channel, and estimator. In Gaussian location experiments, comparison decoders turn Bayes location error into an algorithmic lower bound on a decision-aligned Gaussian range. Conversely, a simultaneous pointwise Gaussian envelope yields an algorithmic upper bound on the same decoding risk. Ordinary nearest-neighbor decoding is the simplest instance; the separating example below uses an unrestricted norm-regularized nearest-neighbor rule over the full ambient class, reflecting the inductive-bias/interpolation principle that good empirical fit must be paired with a locality or complexity bias (Cover and Hart, 1967; Belkin et al., 2018). This conversion is variance-normalized and valid for arbitrary priors, with magnitude controlled by the ghost-mass-induced pointwise dimension. When applied to hard priors localized at the tested radius, a class-supremum relaxation yields a Sudakov-type single-radius obstruction. We then construct an elementary heterogeneous-orthogonal-basis "hub–leaves–cloud" example in which the fixed-prior Fano relaxation is vacuous, the algorithmic ghost-mass lower envelope is sharp for the chosen Bayes problem and for the pointwise upper envelope on the estimator-selected subatlas, while the full-class minimax risk and the global Gaussian supremum scale are much larger.

These results clarify why a Bayesian algorithmic lower bound can provide the right certificate for a pointwise upper envelope under a specified prior, even when the ambient class is overparameterized. In this sense, the pointwise and algorithmic viewpoint developed here is analogous to probabilistic refinements of worst-case analysis in theoretical computer science (Yao, 1977; Levin, 1986; Spielman and Teng, 2004). Just as distributional or smoothed input models can make the effective computational complexity of an algorithm much smaller than its worst-case complexity, a concrete estimator and reference law can make the effective probabilistic or information-theoretic complexity of a rich field much smaller than what is suggested by a global supremum or class-wide relaxation.[1]

The paper then records a finite-cutoff renormalization analogue of this pointwise viewpoint. The analogy with deep feature learning is not simply that layers resemble levels. Rather, the essential common structure is the internal linear *feature/operator inner product* that appears within each composite layer or level. The shared proof architecture is exact finite-step telescoping, a pointwise ellipsoidal metric induced by a learned or renormalized response Gram, and a hierarchical atlas separating local effective dimension from global prior-independence cost. The practical significance in statistical field theory is that RG is fundamentally about controlling coarse marginals after integrating out fluctuations. For globally non-log-concave and anisotropic interacting fields, a direct scalar Lipschitz or global convex-geometric bound is usually too crude. The finite-cutoff viewpoint instead allows one to work on stable local charts where the effective action has a curvature lower bound; directions below the finite resolution are absorbed as approximation error, and the global nonconvexity is paid through an atlas cost. We state this part as an abstract deterministic complexity theorem, together with a chartwise nonconvex example. The graph application later in the paper uses the same local-to-global logic to compare two probabilistic questions: controlling a local-time field on a specified target set versus reducing the problem to global cover time. The phase-star example shows that retaining the target field can give a strictly smaller scale than any argument that first proves full cover.

We keep the limitations explicit. Pointwise complexity provides a useful way to organize finite-cutoff Gaussian and RG geometry, but it should not be viewed as a standalone program for proving Wick factorization, marginal triviality, a universal-approximation-based RG theorem, or the Yang-Mills mass gap. Those problems require substantial model-specific contraction, Gaussianization, or gauge-invariant correlation estimates.

[1] Here "algorithmic" is used in the decision-theoretic sense of bounds depending on a concrete estimator. This is distinct from classical algorithmic information theory, which is traditionally formulated for finite strings or countable description spaces. For related notions of pointwise dimension and their connections to algorithmic and effective statistical dimension, see Li and Xu (2026); Lutz (2016).

Terminology for radii, resolutions, and RG levels. The word "scale" appears in several neighboring literatures, so we keep the terminology separate. The Bayesian lower envelope is a *single-radius* statement at a testing radius Δ, whereas generic chaining is multiresolution and integrates over radii. The Gaussian upper theorem also uses neighborhood radii, such as ε, r_0, and s_0, only as metric resolutions. In the renormalization section we use *RG level* for the index of a block-spin or fluctuation integration, and *finite-cutoff* or *finite-step* for the non-infinitesimal nature of the telescoping argument. Finally, in the separating example the numbers R and S are norms of two different parts of the ambient class. These are related by analogy but are not the same mathematical object.

1.1 Main pointwise Gaussian theorem

Let $\{X_x\}_{x\in T}$ be a centered Gaussian process on a finite index set T. Fix an anchor $x_0 \in T$ such that $X_{x_0} = 0$ almost surely, and define

$$d(x,y) := \left(\mathbb{E}|X_x - X_y|^2\right)^{1/2}, \qquad \sigma(x) := (\mathbb{E}X_x^2)^{1/2} = d(x,x_0), \qquad R_* := \sup_{x\in T} \sigma(x).$$

For an ambient prior $\mu \in \Delta(T)$, where $\Delta(T)$ denotes the set of probability measures on T, define the localized pointwise Fernique–Talagrand functional

$$\Phi_\mu(x) := \int_0^{4\sigma(x)} \sqrt{\log \frac{1}{\mu(B_d(x,\varepsilon))}}\, d\varepsilon, \qquad B_d(x,\varepsilon) := \{y \in T : d(x,y) \le \varepsilon\}. \tag{1.1}$$

Here the upper limit is local: it is proportional to the standard deviation $\sigma(x)$, rather than the global radius. The factor 4 is inessential; it absorbs the two constant-factor losses coming from anchoring and from the nearest-point pushforward used in the ambient-equivalence step. If one instead works with the non-local full-radius integral, then the upper limits R_* and $2R_*$ are equivalent up to universal constants, since the entropy integrand is nonincreasing in ε.

Write

$$\Phi_{\mu,*} := \sup_{x\in T} \Phi_\mu(x), \qquad \log_2^+(u) := \max\{\log_2 u, 0\}.$$

The degenerate case $R_* = 0$ is trivial, and the statements below are understood to be vacuous if $\Phi_{\mu,*} = +\infty$.

For $r_0 > 0$, $s_0 \in (0, R_*]$, and $\delta \in (0,1)$, set

$$M_{r_0,s_0} := \left(1 + \lceil \log_2^+(\Phi_{\mu,*}/r_0)\rceil\right)\left(1 + \lceil \log_2(R_*/s_0)\rceil\right), \tag{1.2}$$

and

$$\mathcal{E}_{\mu,r_0,s_0}(x;\delta) := \max\{\Phi_\mu(x), r_0\} + \max\{\sigma(x), s_0\}\sqrt{\log\left(\frac{eM_{r_0,s_0}}{\delta}\right)}. \tag{1.3}$$

Thus all logarithmic losses from the two successive pointwise-peeling steps are isolated in M_{r_0,s_0}.

We state the main result as a variance-aware pointwise Gaussian majorizing-measure theorem in high probability.

Theorem 1.1 (Pointwise Gaussian majorizing-measure bound)**.** *Assume $0 < R_* < \infty$ and $\Phi_{\mu,*} < \infty$. There exists an absolute constant $C > 0$ such that, for every ambient prior $\mu \in \Delta(T)$, every $r_0 > 0$, every $s_0 \in (0, R_*]$, and every $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\forall x \in T : \qquad X_x \le C\, \mathcal{E}_{\mu,r_0,s_0}(x;\delta). \tag{1.4}$$

Consequently, applying the same bound to X and $-X$ and taking a union bound, with probability at least $1-\delta$,

$$\forall x \in T: \qquad |X_x| \le A_{\mu,r_0,s_0}(x;\delta), \qquad A_{\mu,r_0,s_0}(x;\delta) := C\,\mathcal{E}_{\mu,r_0,s_0}(x;\delta/2). \tag{1.5}$$

When r_0, s_0 are fixed, we abbreviate A_{μ,r_0,s_0} to A_μ.

Besides providing the Gaussian-process counterpart of the pointwise empirical-process bound in Li and Xu (2026, Theorem1), the present result makes a key technical refinement: the global boundedness parameter for the metric is replaced by the local variance scale $4\sigma(x)$. Moreover, in the Gaussian-process setting we do not need the advanced symmetrization machinery or the mixed empirical-ghost metric used in Li and Xu (2026, Theorem1).

Before presenting the in-expectation consequence, we record the corresponding anchored form of the classical generic-chaining scale. This is a localized reformulation of the usual FerniqueTalagrand theorem: since the cutoff $4\sigma(x)$ is not the standard full-radius presentation, the global radius R_* must remain part of the anchored scale.

Proposition 1.2 (Classical scale in anchored local form)**.** *There exist absolute constants $0 < c < C < \infty$ such that*

$$c\left(R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}\right) \le \mathbb{E}\sup_{x\in T} X_x \le C\left(R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}\right). \tag{1.6}$$

Moreover, since the process is centered and anchored,

$$\mathbb{E}\sup_{x\in T}|X_x| \asymp \mathbb{E}\sup_{x\in T} X_x.$$

Therefore

$$\mathbb{E}\sup_{x\in T}|X_x| \asymp R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}. \tag{1.7}$$

We now record the in-expectation consequence of Theorem 1.1. In particular, after optimizing over the ambient prior, the resulting global in-expectation bound is sharp up to universal constants. The variance-radius term is not an additional loss: it is part of the anchored generic-chaining scale in (1.7), while the peeling logarithms disappear by taking $r_0 \ge \Phi_{\mu,*}$ and $s_0 = R_*$.

Corollary 1.3 (In-expectation consequence and sharp global scale)**.** *Under the assumptions of Theorem 1.1, there exists an absolute constant $C > 0$ such that*

$$\mathbb{E}\left[\sup_{x\in T} \frac{\big(|X_x| - C\max\{\Phi_\mu(x), r_0\}\big)_+}{\max\{\sigma(x), s_0\}}\right] \le C\sqrt{\log(eM_{r_0,s_0})}. \tag{1.8}$$

Consequently, for every $x \in T$,

$$\mathbb{E}|X_x| \le C\max\{\Phi_\mu(x), r_0\} + C\max\{\sigma(x), s_0\}\sqrt{\log(eM_{r_0,s_0})}. \tag{1.9}$$

At the global level, taking $r_0 \ge \Phi_{\mu,*}$ and $s_0 = R_*$ gives $M_{r_0,s_0} = 1$, hence

$$\mathbb{E}\sup_{x\in T}|X_x| \le C(r_0 + R_*). \tag{1.10}$$

Letting $r_0 \downarrow \Phi_{\mu,*}$, then optimizing over μ, and using (1.7), we obtain the sharp equivalence

$$\mathbb{E}\sup_{x\in T}|X_x| \asymp R_* + \inf_{\mu\in\Delta(T)} \Phi_{\mu,*}. \tag{1.11}$$

Thus the optimized expectation version is optimal up to absolute constants.

Remark (variance term versus chaining complexity). The term in Theorem 1.1 involving $\sigma(x)$ is a variance-sensitive Gaussian tail term. It is necessary for high-probability control, already for a single non-anchor point $X_x \sim N(0, \sigma(x)^2)$, but it is not a replacement for the chaining complexity $\Phi_\mu(x)$, which carries the local geometry of the index set. For a fixed marginal, $\mathbb{E}|X_x| = \sqrt{\frac{2}{\pi}}\,\sigma(x)$, so the extra factor $\sqrt{\log(eM_{r_0,s_0})}$ in (1.9) is not intrinsic to $\mathbb{E}|X_x|$. It is the price of extracting a simultaneous pointwise envelope by peeling first over the local Fernique–Talagrand complexity and then over the local variance. At the global in-expectation level, this peeling factor disappears, yielding the sharp scale in (1.11).

Remark (pointwise high probability). For Gaussian processes, subset-level in-expectation bounds transfer to high-probability bounds through Borell–Tsirelson concentration, with variance scale $\sigma_H = \sup_{x \in H}(\mathbb{E}X_x^2)^{1/2}$ for a fixed subset H. The genuinely pointwise statement is not obtained from this transfer alone: after the subset-homogeneous estimate is established, one first applies uniform pointwise peeling to localize the Fernique–Talagrand term $\Phi_\mu(x)$, and then applies the same peeling mechanism a second time to improve the tail scale from the global radius to the local variance $\sigma(x)$. Conversely, the resulting high-probability envelope implies the stated in-expectation consequence by integrating the tail.

1.2 Main Bayesian algorithmic lower bounds and consequences

The pointwise upper envelope is frequentist, simultaneous, multiscale, and pathwise in the Gaussian field. A lower-bound analogue should therefore be stated with care. We first record a one-scale Bayesian and estimator-dependent primitive. Let (T, d) be a finite metric space, let $\pi \in \Delta(T)$ be a known prior, and let $\{P_t : t \in T\}$ be an experiment on an observation space $\mathcal{Y}$. For an estimator $\widehat{t} : \mathcal{Y} \to T$, a reference law $Q \in \Delta(\mathcal{Y})$, and a scale $\Delta > 0$, define

$$\rho_{\Delta,Q}(\widehat{t}) := \mathbb{P}_{t\sim\pi,\, Y\sim Q}\{d(t, \widehat{t}(Y)) < \Delta\} = \mathbb{E}_{Y\sim Q}\, \pi\big(B_d(\widehat{t}(Y), \Delta)\big)\,,$$

and

$$\mathcal{I}_\pi(Q) := \mathbb{E}_{t\sim\pi} D_{\mathrm{KL}}(P_t \| Q).$$

The following result is the Gaussian-process and KL specialization of the interactive Fano method of Chen et al. (2024, Theorem 2). The original theorem provides a unifying information-theoretic principle for proving lower bounds in learning and decision-making problems.

Proposition 1.4 (Bayesian algorithmic lower envelope)**.** *For every estimator $\widehat{t}$, every $\Delta > 0$, and every reference law Q with $0 < \rho_{\Delta,Q}(\widehat{t}) < 1$,*

$$\mathbb{P}_{t\sim\pi,\, Y\sim P_t}\{d(t, \widehat{t}(Y)) \geq \Delta\} \geq \left(1 - \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/\rho_{\Delta,Q}(\widehat{t}))}\right)_+ . \tag{1.12}$$

Consequently,

$$\mathbb{E}_{t\sim\pi,\, Y\sim P_t} d(t, \widehat{t}(Y)) \geq \sup_{\Delta>0,\, Q} \Delta \left(1 - \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/\rho_{\Delta,Q}(\widehat{t}))}\right)_+ . \tag{1.13}$$

The key point is that $\rho_{\Delta,Q}(\widehat{t})$ is not immediately replaced by the class-wide quantity

$$q_\pi(\Delta) := \sup_{a \in T} \pi(B_d(a, \Delta)).$$

Keeping the exact ghost mass $\mathbb{E}_{Y\sim Q}\pi(B_d(\widehat{t}(Y),\Delta))$ makes the lower bound algorithmic: it is attached to the image of the actual estimator under ghost data. Replacing it by $q_\pi(\Delta)$ gives the class-wide small-ball term used in classical Fano-type lower bounds (Zhang, 2006) and in the fractional-covering specialization of Chen et al. (2024). In the Gaussian nearest-neighbor comparison below, this small-ball term must still be balanced against the information radius of the Gaussian channel.

We next present a lower-bound comparison for the Gaussian process itself, rather than only for its induced distance. The result is stated for a comparison decoder. This includes ordinary nearest neighbor and the norm-regularized rule used in the separating example below.

Theorem 1.5 (Algorithmic lower bound for comparison decoders)**.** *Let $v : T \to \mathcal{H}$ be an embedding into a finite-dimensional Hilbert space and set $d(s,t) = \|v_s - v_t\|_{\mathcal{H}}$. Let*

$$Y = v_\Theta + \tau Z, \qquad \Theta \sim \pi, \qquad Z \sim N(0, I_{\mathcal{H}}),$$

and let $m \in \mathcal{H}$ be any reference center. Set

$$Q_{\tau,m} := N(m, \tau^2 I_{\mathcal{H}}), \qquad \mathcal{I}_{\pi,m} := \frac{1}{2\tau^2}\int_T \|v_t - m\|^2\,\pi(dt).$$

With $\bar{v}_\pi := \int_T v_t\,\pi(dt)$, the centered choice $m = \bar{v}_\pi$ gives

$$\mathcal{I}_{\pi,\bar{v}_\pi} = \frac{\mathsf{V}_\pi}{4\tau^2}, \qquad \mathsf{V}_\pi := \iint d(s,t)^2\pi(ds)\pi(dt).$$

Let $\widehat{\Theta}(Y)$ be an estimator with values in T. Assume that there is a penalty $\Omega : T \to \mathbb{R}$ such that, under the true joint law,

$$\|Y - v_{\widehat{\Theta}}\|^2 + \Omega(\widehat{\Theta}) \le \|Y - v_\Theta\|^2 + \Omega(\Theta), \qquad \Omega(\widehat{\Theta}) \ge \Omega(\Theta) \quad a.s. \tag{1.14}$$

If, for some $c \in (0,1)$ and $\Delta > 0$,

$$\mathcal{I}_{\pi,m} + \log 2 \le (1-c)\log\frac{1}{\rho_{\Delta,Q_{\tau,m}}(\widehat{\Theta})}, \tag{1.15}$$

then

$$\mathbb{E}_{\Theta,Y} d(\Theta, \widehat{\Theta}(Y)) \ge c\Delta, \tag{1.16}$$

and, for the Gaussian process $X_t := \langle Z, v_t\rangle$ built from the same Gaussian vector Z,

$$\mathbb{E}_{\Theta,Z}\big[X_{\widehat{\Theta}} - X_\Theta\big] \ge \frac{c^2\Delta^2}{2\tau}. \tag{1.17}$$

Thus the exact ghost mass gives an algorithmic lower bound on the Gaussian range sampled by the comparison decoder.

Ordinary nearest neighbor corresponds to $\Omega \equiv 0$. The unrestricted norm-regularized nearest-neighbor rule

$$\widehat{\Theta}_\lambda(Y) \in \arg\min_{a\in T}\big\{\|Y - v_a\|^2 + \lambda\|v_a\|^2\big\}$$

is also covered whenever the realized outputs have norm at least the norm of the true parameter, as in the separating example below. This is a canonical minimum-complexity inductive bias consist with modern perspectives, rather than an explicit restriction of the hypothesis class.

The same pathwise comparison underlying Theorem 1.5, together with the pointwise Gaussian upper envelope, gives an algorithmic upper bound for Gaussian location decoding. Thus, when the two estimates match, the Bayesian algorithmic lower bound and the pointwise upper envelope certify the sharpness of one another.

Corollary 1.6 (Algorithmic pointwise upper bound for estimator decoding)**.** *Work in the Gaussian location setup of Theorem 1.5. Thus T is embedded as $\{v_t : t \in T\}$ in a Hilbert space,*

$$d(s,t) = \|v_s - v_t\|, \qquad Y = v_\Theta + \tau Z, \qquad X_t := \langle Z, v_t \rangle,$$

where Z is standard Gaussian. Let $\widehat{\Theta} = \widehat{\Theta}(Y)$ be a decoder satisfying the comparison condition (1.14); in particular, the condition implies

$$\|Y - v_{\widehat{\Theta}(Y)}\|^2 \le \|Y - v_\Theta\|^2 \tag{1.18}$$

under the joint law of (Θ, Y).

Suppose that a deterministic envelope $A : T \to [0,\infty)$ satisfies

$$\mathbb{P}_Z\big\{\forall t \in T : \ |X_t| \le A(t)\big\} \ge 1 - \delta. \tag{1.19}$$

Then, with probability at least $1-\delta$ under the joint law of (Θ, Y),

$$d(\Theta, \widehat{\Theta}(Y))^2 \le 2\tau\big(A(\Theta) + A(\widehat{\Theta}(Y))\big). \tag{1.20}$$

Consequently, if

$$D_T := \sup_{s,t\in T} d(s,t) < \infty,$$

then

$$\mathbb{E}d(\Theta, \widehat{\Theta}(Y)) \le \mathbb{E}\sqrt{2\tau\big(A(\Theta) + A(\widehat{\Theta}(Y))\big)} + \delta D_T. \tag{1.21}$$

In particular, after adjoining an anchor point t_0 with $v_{t_0} = 0$ and $X_{t_0} = 0$ if necessary, one may take

$$A(t) = A_{\mu,r_0,s_0}(t;\delta)$$

from (1.5) in Theorem 1.1, with the prior extended to $T \cup \{t_0\}$. This yields a pointwise-complexity upper bound for the estimation risk.

We defer to Section 3.2 a route from the algorithmic bound to a Sudakov-type comparison via supremum relaxation, which consists of several technical results.

We would like to clarify that the Bayesian algorithmic lower bounds in the form of Proposition 1.4 and Theorem 1.5, for example in the nearest-neighbor formulation, has a different scope from traditional Bayesian lower bounds. The prior is used as a device for averaging and comparison, but the object being certified is always the performance of a concrete algorithm in the pointwise frequentist setting, rather than the behavior of an unknown parameter drawn from the prior.

Why keep the Bayesian and algorithmic form. The distinction clarifies the role of minimax reasoning. If the goal is Bayes risk under a particular known prior π, then a benchmark of the form

$$\inf_{\widehat{t}} \sup_{t\in T} R(t, \widehat{t})$$

may be poorly aligned with the problem: it asks for performance against the worst point of T, whereas the Bayesian problem asks for performance under the given prior geometry. Conversely, maximizing over priors can erase the structure that makes a particular algorithm appropriate. In this sense, the algorithm-dependent ghost mass in $\rho_{\Delta,Q}(\widehat{t})$ records how the actual estimator maps ghost observations into metric balls. No single algorithm can be uniformly best for all possible priors or all possible geometries, so a well-specified Bayesian lower envelope can be a more informative optimality certificate for a pointwise frequentist upper envelope than a purely class-wide minimax statement. The algorithmic form is therefore crucial: it allows us to evaluate a fixed estimator under specified prior geometries for the purpose of pointwise comparison.

This perspective also suggests why simply restricting the hypothesis class to a structured class is not always a satisfactory replacement. First, in the pointwise-complexity theory of Li and Xu (2026), low compression complexity depends on both the realized hypothesis and the data distribution; it is therefore a data-centric, pointwise property rather than merely membership in a fixed combinatorial subclass. Second, computationally efficient procedures may not output elements of the exact compression class. The familiar lesson from sparse regression is that tractable algorithms often work through relaxations or implicit regularization rather than by solving the exact combinatorial model-selection problem. The algorithmic lower envelope is designed for this regime: it evaluates the lower-bound obstruction through the estimator actually used, under a prior that represents the intended geometry.

Relationship to the pointwise upper theorem and existing lower bounds. The Bayesian algorithmic lower bound also clarifies the relation to the pointwise upper theorem. The upper envelope is frequentist, simultaneous, multiscale, and high-probability: it is pathwise in the Gaussian field and integrates local prior masses across scales. By contrast, the lower bound is Bayesian, one-scale, and estimator-dependent: it averages over $\Theta \sim \pi$ and $Y \sim P_\Theta$ for a fixed estimator, while the additional ghost/reference law $Y \sim Q$ enters through the quantile term $\rho_{\Delta,Q}(\widehat{t})$. This parallels the information-theoretic upper–lower duality emphasized by Zhang (2006), but with the additional point that the lower bound is a data-processing statement valid for every estimator.

DEC-type lower bounds (Foster et al., 2021, 2023) provide an independent motivation for incorporating algorithmic dependence into the fundamental limits of online regret, exploration, and interactive decision making. This perspective further motivates the Bayesian algorithmic lower bound of Chen et al. (2024) and related variants (Xu and Zeevi, 2025b). The present algorithmic lower-bound perspective isolates a distinct but complementary principle: a prior-mass lower-bound analogue of pointwise complexity, in the spirit of the pointwise generalization framework of Li and Xu (2026) for deep representation learning.

1.3 Separating example construction and pointwise certification

Besides the pointwise upper envelope and Bayesian algorithmic lower bounds, a central contribution of the paper is the construction of a separating example that illustrates how these two mechanisms certify local-geometric optimality.

Section 4 gives a finite-dimensional heterogeneous-orthogonal-basis separating example showing that the Bayesian algorithmic lower envelope is the correct certificate for a pointwise upper envelope under a specified prior and estimator, even when the ambient class is deliberately overparameterized. The construction has a lower-norm hub–leaf subatlas

$$H_M = \{Rg_0, Re_1, \ldots, Re_M\}$$

and a farther orthogonal cloud

$$C_N = \{Sf_1, \ldots, Sf_N\}, \qquad T_{M,N} = H_M \cup C_N,$$

with a nonuniform prior supported only on the decision-relevant subatlas, $\pi(Rg_0) = 1/2$ and $\pi(Re_i) = 1/(2M)$. We use the unrestricted norm-regularized decoder

$$\widehat{\Theta}_\lambda(Y) \in \arg\min_{a \in T_{M,N}} \{\|Y - a\|^2 + \lambda\|a\|^2\},$$

with a fixed numerical λ, so the learner is not restricted to H_M; rather, the minimum-norm inductive bias selects this subatlas under the relevant ghost law. The example proves the simultaneous separation

$$q_\pi(\Delta) = \sup_a \pi(B_d(a, \Delta)) = \frac{1}{2}, \qquad \rho_{\Delta,Q_\tau}(\widehat{\Theta}_\lambda) \lesssim M^{-1},$$

so the usual fixed-prior Fano relaxation is vacuous while the exact ghost mass has entropy of order $\log M$. Consequently the actual Bayes risk of the decoder is sharp,

$$\mathbb{E}_{\Theta,Y} d(\Theta, \widehat{\Theta}_\lambda(Y)) \asymp R,$$

whereas the full-class minimax risk is governed by the cloud and satisfies

$$\inf_{\widehat{\theta}} \sup_{t \in T_{M,N}} \mathbb{E}_t d(t, \widehat{\theta}(Y)) \asymp S, \qquad S/R \to \infty.$$

For the associated Gaussian process $X_t = \langle Z, t \rangle$, the algorithmically induced certificate

$$\mathcal{G}_{\text{alg}} := \mathbb{E}_{\Theta,Y}\left[X_{\widehat{\Theta}_\lambda(Y)} - X_\Theta\right]$$

matches the pointwise Gaussian scale on the selected subatlas,

$$\mathcal{G}_{\text{alg}} \asymp \mathbb{E} \sup_{s,u \in H_M} (X_s - X_u) \asymp R\sqrt{\log M},$$

while the global Gaussian scale over the full ambient class can be much larger,

$$\mathbb{E} \sup_{s,u \in T_{M,N}} (X_s - X_u) \gtrsim S\sqrt{\log N} \gg R\sqrt{\log M}.$$

Thus the same example separates fixed-prior worst-case Fano, Bayesian algorithmic lower bounds, full-class minimax risk, pointwise Gaussian envelopes, and global Sudakov scale. Its purpose is to demonstrate that pointwise upper bounds and algorithmic lower bounds are naturally decision-aligned: the relevant complexity is the geometry actually selected by the prior and estimator, not necessarily the worst-case geometry of the full ambient class. Together, these separation results clarify why a Bayesian algorithmic lower bound can provide the right certificate for a pointwise upper envelope under a specified prior, even when the ambient class is overparameterized.

1.4 Main finite-cutoff renormalization theorem

We next state the finite-dimensional renormalization theorem in the form used later. Its main role is to provide an analogue of the pointwise complexity theory for deep neural networks developed in Li and Xu (2026). This analogy relies not only on the correspondence between composite layers and RG levels, but more essentially on the common *feature/operator inner-product* structure that appears within each composite layer or level. Let a depth-J RG trajectory be parametrized by level variables

$$g = (g_0, \ldots, g_{J-1}), \qquad g_j \in B_2(R_j) \subset \mathbb{R}^{p_j},$$

and let $S_J(g)$ denote the final effective action, or any finite-dimensional observable extracted from the final effective action, equipped with a semimetric ρ. The theorem is conditional on a finite-scale domination statement of the form

$$\rho(S_J(g'), S_J(g))^2 \le \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g, \varepsilon)(g'_j - g_j) \tag{1.22}$$

for all g' in a finite ρ-neighborhood of g, where the secant Gram G_j are defined via the renormalized operator for exponential-family RG steps in Section 5. This defines the pointwise ellipsoidal RG metric

$$\rho_{G,g,\varepsilon}(g, g')^2 := \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g, \varepsilon)(g'_j - g_j),$$

which is the RG analogue of the pointwise ellipsoidal metric induced by learned feature Gram matrices in the DNN theory. The matrices $G_j(g, \varepsilon)$ are not infinitesimal Hessians; they are finite-scale secant response Grams produced by replacing one RG level at a time.

For a positive semidefinite matrix G, radius R, and resolution $u > 0$, define

$$r_{\text{eff}}(G, R, u) := \max\{k : \lambda_k(G)R^2 \ge u^2\}, \qquad d_{\text{ell}}(G, R, u) := \frac{1}{2} \sum_{k=1}^{r_{\text{eff}}} \log\left(\frac{CR^2\lambda_k(G)}{u^2}\right), \tag{1.23}$$

with eigenvalues in nonincreasing order. Let $V_j(g, \varepsilon)$ be the active eigenspace of $G_j(g, \varepsilon)$, and let $\mathfrak{A}_j(g, \varepsilon, u_j)$ be the atlas cost of choosing the phase label, effective rank, and a reference subspace close to $V_j(g, \varepsilon)$. A concrete Grassmannian expression is given in Section 5.

Theorem 1.7 (Finite-scale RG pointwise dimension, main form)**.** *Assume the finite-scale domination (1.22). Choose numbers $u_j > 0$ such that $\sum_j u_j^2 \le c\varepsilon^2$. Then there exists a data-independent hierarchical prior Π over phase labels, active ranks, reference subspaces, and local scale couplings such that*

$$\log \frac{1}{\Pi(B_\rho(g, \varepsilon))} \le C \sum_{j=0}^{J-1} \left[d_{\text{ell}}(G_j(g, \varepsilon), R_j, u_j) + \mathfrak{A}_j(g, \varepsilon, u_j)\right]. \tag{1.24}$$

Thus finite-cutoff renormalization has the same complexity structure as nonlinear feature-learning DNN: a local ellipsoidal effective dimension plus a global atlas cost at each scale.

We then apply the theorem to a nonconvex phase-atlas construction, showcasing how finite-cutoff geometry can control coarse marginals of globally non-log-concave and anisotropic measures. The theorem and the example should be read as abstract finite-cutoff results, not as a shortcut to continuum constructive QFT. To prove a scaling-limit theorem such as four-dimensional marginal triviality, one also needs a model-specific contraction or Gaussianization estimate for non-Gaussian connected correlations. The pointwise-dimension theorem supplies the local finite-scale complexity layer once such RG or correlation estimates are available.

1.5 Main graph local-time application

We give a specialized application of the pointwise-complexity principle to the control of local times in Gaussian free fields. The results separate two distinct ways of using Gaussian information, highlighting the advantage of a field-level route. In the field-level route, one fixes a target set, controls the pinned GFF on that set by a pointwise envelope, and then transfers the resulting estimate through the RayKnight theorem to a local-time statement. In the global route, one first establishes that the entire graph has been covered and only afterwards deduces that the target set has been visited. These two stopping-time arguments may occur at different scales.

Section 6.1 proves this separation on phase-star hierarchies. A target subatlas I is encoded by an atlas prior q, and the same ambient pointwise envelope can be evaluated on the selected target. The exact law

$$\mathbb{P}\{\tau_I \le \tau(t)\} = (1 - e^{-t})^{|I|}, \qquad \mathbb{P}\{\tau_{\mathrm{cov}} \le \tau(t)\} = (1 - e^{-t})^N$$

shows that target cover has intrinsic inverse-root-local-time scale $\log |I|$, whereas any route through full cover pays $\log N$. The pointwise envelope pays the target code length $\log(|I|/q(I))$, and the chart-decorated phase-star adds a compressed within-chart term. This gives a concrete finite-graph analogue of the phase/subspace atlas cost in Section 5.

1.6 Organization

Section 2 proves the pointwise Gaussian theorem. Section 3 proves the Bayesian algorithmic lower bounds and their consequences. Section 4 construct an important heterogeneous-cube example separating the fixed-prior Fano relaxation, the algorithmic ghost-mass lower bound, the pointwise Gaussian envelope, and the full-class minimax/global Gaussian scale. Section 5 develops the finite-cutoff renormalization pointwise-complexity theory, including the DNN–RG dictionary, secant-Gram metric domination, and phase/subspace atlas covering. Section 6 gives the graph and local-time applications. Section 7 summarizes the pointwise/algorithmic scope and the main lessons.

2 Proof of the Pointwise Gaussian Theorem

The proof follows a subset-homogeneity and peeling strategy. First, for each fixed subset $H \subseteq T$, we combine an anchored form of Fernique's majorizing-measure upper bound (Fernique, 1975; Talagrand, 1987), the ambient equivalence principle of pointwise dimension, and Borell–Tsirelson concentration. Second, we apply the one-parameter uniform pointwise peeling lemma twice: first to localize the Fernique–Talagrand integral $\Phi_\mu(x)$, and then to localize the variance scale from R_* to $\sigma(x)$. This avoids introducing a separate two-dimensional peeling corollary.

The following uniform pointwise peeling lemma was originally developed for empirical processes in Xu and Zeevi (2025a). It identifies minimal conditions for obtaining pointwise envelopes, up to negligible double-logarithmic factors, without imposing unrealistic Bernstein-type or rigid sub-root assumptions. Necessary and sufficient justifications and discussions of data dependence are provided in Li and Xu (2026). Here we extend the lemma to general indexed processes, including Gaussian processes.

Lemma 2.1 (Uniform pointwise peeling for indexed processes)**.** *Let $\{Z_x\}_{x\in T}$ be random variables, let $d_0 : T \to [0, R]$, and let $\psi(r; \delta)$ be nondecreasing in r and nonincreasing in δ. Assume that for every fixed subset $H \subseteq T$ and every $\delta \in (0, 1)$,*

$$\mathbb{P}\left(\sup_{x\in H} Z_x \le \sup_{x\in H} \psi(d_0(x); \delta)\right) \ge 1 - \delta. \tag{2.1}$$

Then for every $r_0 > 0$, with probability at least $1 - \delta$, uniformly over all $x \in T$,

$$Z_x \le \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{1 + \lceil \log_2^+(R/r_0)\rceil}\right), \tag{2.2}$$

where $\log_2^+(u) = \max\{\log_2 u, 0\}$.

Proof. Let $m := \lceil \log_2^+(R/r_0) \rceil$. The sets

$$H_0 := \{x : d_0(x) \leq r_0\}, \qquad H_j := \{x : 2^{j-1} r_0 < d_0(x) \leq 2^j r_0\} \quad (1 \leq j \leq m)$$

cover T. Apply (2.1) to each H_j with confidence level $\delta/(m+1)$, and take a union bound. On the resulting event, if $x \in H_0$, then

$$Z_x \leq \psi\left(r_0; \frac{\delta}{m+1}\right) \leq \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{m+1}\right),$$

and if $x \in H_j$ for $j \geq 1$, then $2^j r_0 \leq 2d_0(x)$, so

$$Z_x \leq \psi\left(2^j r_0; \frac{\delta}{m+1}\right) \leq \psi\left(\max\{2d_0(x), r_0\}; \frac{\delta}{m+1}\right).$$

This proves the claim. $\square$

We also use the following localized version of the ambient equivalence principle of pointwise dimension from Lemma 7 of Li and Xu (2026).

Lemma 2.2 (Ambient equivalence for the localized integral)**.** *Let (T, d) be a finite metric space and let $H \subseteq T$. Let $p : T \to H$ be a nearest-point selector, so that $d(y, p(y)) = \min_{h \in H} d(y, h)$, and let $\mu_H = p_\# \mu$ be the pushforward of an ambient prior $\mu \in \Delta(T)$. Then, for every $x \in H$ and every $\varepsilon > 0$,*

$$\mu_H(B_d(x, 2\varepsilon)) \geq \mu(B_d(x, \varepsilon)).$$

Consequently, for the localized functional Φ in (1.1),

$$\Phi_{\mu_H}(x) \leq 2\Phi_\mu(x), \qquad x \in H. \tag{2.3}$$

Proof. If $y \in B_d(x, \varepsilon)$, then

$$d(p(y), x) \leq d(p(y), y) + d(y, x) \leq 2d(y, x) \leq 2\varepsilon,$$

because $x \in H$ and $p(y)$ is a nearest point in H. Hence $p(B_d(x, \varepsilon)) \subseteq B_d(x, 2\varepsilon)$, which gives the ball-mass inequality. Therefore

$$\Phi_{\mu_H}(x) = \int_0^{4\sigma(x)} \sqrt{\log \frac{1}{\mu_H(B_d(x, u))}}\, du \leq 2 \int_0^{2\sigma(x)} \sqrt{\log \frac{1}{\mu(B_d(x, \varepsilon))}}\, d\varepsilon \leq 2\Phi_\mu(x),$$

where we used the change of variables $u = 2\varepsilon$. $\square$

Lemma 2.3 (Anchored subset majorizing-measure bound)**.** *For every fixed subset $H \subseteq T$, every ambient prior $\mu \in \Delta(T)$, and*

$$\sigma_H := \sup_{x \in H} \sigma(x),$$

there is an absolute constant $C > 0$ such that

$$\mathbb{E} \sup_{x \in H} X_x \leq C \sup_{x \in H} \Phi_\mu(x) + C\sigma_H. \tag{2.4}$$

Proof. Let $H_0 := H \cup \{x_0\}$, and let $p : T \to H_0$ be a nearest-point selector. Define the probability measure

$$\bar{\mu}_H := \frac{1}{2}\delta_{x_0} + \frac{1}{2}p_{\#}\mu$$

on H_0. Fernique's majorizing-measure upper bound (Fernique, 1975; Talagrand, 1987), applied to the restricted process on H_0, gives

$$\mathbb{E}\sup_{x\in H} X_x \le C \sup_{x\in H_0}\int_0^{2\sigma_H}\sqrt{\log\frac{1}{\bar{\mu}_H(B_d(x,u))}}\,du,$$

since $\operatorname{diam}(H_0,d) \le 2\sigma_H$. For $x = x_0$, the integral is at most $2\sigma_H\sqrt{\log 2}$, because $\bar{\mu}_H\{x_0\} \ge 1/2$. For $x \in H$, split the integral at $4\sigma(x)$. On $[4\sigma(x), 2\sigma_H]$, the ball $B_d(x,u)$ contains x_0, hence has $\bar{\mu}_H$-mass at least $1/2$, so this part is bounded by $C\sigma_H$. On $[0, 4\sigma(x)]$, Lemma 2.2 and the extra factor $1/2$ in $\bar{\mu}_H$ give a bound of order $\Phi_\mu(x) + \sigma(x)$. Taking the supremum over $x \in H$ proves (2.4). □

Lemma 2.4 (Subset-homogeneous Gaussian estimate)**.** *For every fixed subset $H \subseteq T$, every ambient prior $\mu \in \Delta(T)$, and every $\delta \in (0,1)$,*

$$\mathbb{P}\left(\sup_{x\in H} X_x \le C\sup_{x\in H}\Phi_\mu(x) + C\sigma_H\sqrt{\log(e/\delta)}\right) \ge 1-\delta, \tag{2.5}$$

where $\sigma_H = \sup_{x\in H}\sigma(x)$.

Proof. By Lemma 2.3,

$$\mathbb{E}\sup_{x\in H} X_x \le C\sup_{x\in H}\Phi_\mu(x) + C\sigma_H.$$

Realize $X_x = \langle g, v_x\rangle$ as an isonormal process on a Hilbert space. The map $g \mapsto \sup_{x\in H}\langle g, v_x\rangle$ is σ_H-Lipschitz. Borell–Tsirelson concentration therefore gives the desired inequality, after absorbing the expectation term $C\sigma_H$ into $C\sigma_H\sqrt{\log(e/\delta)}$. □

Proof of Theorem 1.1. Fix an arbitrary subset $H \subseteq T$. Applying Lemma 2.1 on the index set H, with $Z_x = X_x$, $d_0(x) = \Phi_\mu(x)$, $R = \Phi_{\mu,*}$, and using the subset estimate (2.5), yields the following: for every $r_0 > 0$ and every $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\forall x \in H: \qquad X_x \le C\max\{\Phi_\mu(x), r_0\} + C\sigma_H\sqrt{\log\left(\frac{eM_\Phi}{\delta}\right)}, \tag{2.6}$$

where

$$M_\Phi := 1 + \lceil \log_2^+(\Phi_{\mu,*}/r_0)\rceil.$$

Since H was arbitrary, (2.6) is now a subset-homogeneous estimate for the centered residual

$$Y_x := X_x - C\max\{\Phi_\mu(x), r_0\}$$

with complexity $\sigma(x)$. Applying Lemma 2.1 a second time, now with $Z_x = Y_x$, $d_0(x) = \sigma(x)$, $R = R_*$, and

$$\psi(s;\delta) := Cs\sqrt{\log\left(\frac{eM_\Phi}{\delta}\right)},$$

gives, with probability at least $1-\delta$,

$$\forall x \in T: \qquad Y_x \le C \max\{\sigma(x), s_0\} \sqrt{\log\left(\frac{eM_\Phi M_\sigma}{\delta}\right)},$$

where

$$M_\sigma := 1 + \lceil \log_2(R_*/s_0) \rceil.$$

Combining the last display with the definition of Y_x proves (1.4), since $M_{r_0,s_0} = M_\Phi M_\sigma$. The absolute-value estimate (1.5) follows by applying the same argument to $-X$ and taking a union bound. □

Proof of Proposition 1.2. The upper bound follows from Lemma 2.3 with $H = T$, followed by optimization over μ. For the lower bound, choose x_* with $\sigma(x_*) = R_*$. Since $X_{x_0} = 0$,

$$\sup_{x \in T} X_x \ge (X_{x_*})_+, \qquad \mathbb{E}(X_{x_*})_+ = \frac{R_*}{\sqrt{2\pi}}.$$

It remains to lower-bound the optimized local functional. Let

$$\Phi_\mu^{\text{full}}(x) := \int_0^{2R_*} \sqrt{\log \frac{1}{\mu(B_d(x,\varepsilon))}}\, d\varepsilon.$$

The classical majorizing-measure theorem gives

$$\inf_{\mu \in \Delta(T)} \sup_x \Phi_\mu^{\text{full}}(x) \lesssim \mathbb{E} \sup_{x \in T} X_x.$$

Since $\Phi_\mu(x) \le \Phi_\mu^{\text{full}}(x)$ for every x, we have

$$\inf_\mu \Phi_{\mu,*} \le \inf_\mu \sup_x \Phi_\mu^{\text{full}}(x) \lesssim \mathbb{E} \sup_{x \in T} X_x.$$

Together with the radius lower bound this proves the left side of (1.6). The absolute-value statement follows from

$$\sup_x X_x \le \sup_x |X_x| \le \sup_x X_x + \sup_x(-X_x)$$

and symmetry. □

Proof of Corollary 1.3. Let

$$Y := \sup_{x \in T} \frac{\big(|X_x| - C \max\{\Phi_\mu(x), r_0\}\big)_+}{\max\{\sigma(x), s_0\}}.$$

By the two-sided form of Theorem 1.1, after increasing C if necessary,

$$\mathbb{P}\left(Y > C\sqrt{\log\left(\frac{eM_{r_0,s_0}}{\delta}\right)}\right) \le \delta, \qquad \delta \in (0,1).$$

Integrating this tail gives

$$\mathbb{E}Y \le C\sqrt{\log(eM_{r_0,s_0})},$$

which proves (1.8). The pointwise bound (1.9) follows by dropping the supremum, and the global bound (1.10) follows by taking $r_0 \ge \Phi_{\mu,*}$, $s_0 = R_*$, and then letting $r_0 \downarrow \Phi_{\mu,*}$. Optimizing over μ and applying Proposition 1.2 proves (1.11). □

3 Proofs and Consequences of the Bayesian algorithmic lower bounds

3.1 Proof of the Bayesian algorithmic lower and upper bounds

Proof of Proposition 1.4. The result is the Gaussian-process and KL specialization of the interactive Fano method of Chen et al. (2024, Theorem 2); we give a self-contained proof. Fix $\widehat{t}$, Q, and Δ, and let

$$E := \{(t, Y) : d(t, \widehat{t}(Y)) < \Delta\}.$$

Under the true joint law $P := \int \pi(dt) P_t(dY)$, write $p = P(E)$. Under the ghost law $P_0 := \pi \otimes Q$, write

$$q = P_0(E) = \rho_{\Delta,Q}(\widehat{t}).$$

By data processing,

$$\mathcal{I}_\pi(Q) = D_{\mathrm{KL}}(P \| P_0) \geq D_{\mathrm{KL}}(\mathrm{Bern}(p) \| \mathrm{Bern}(q)).$$

The elementary inequality

$$D_{\mathrm{KL}}(\mathrm{Bern}(p) \| \mathrm{Bern}(q)) \geq p \log \frac{1}{q} - \log 2$$

gives

$$p \leq \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/q)}.$$

Therefore

$$P\{d(t, \widehat{t}(Y)) \geq \Delta\} = 1 - p \geq 1 - \frac{\mathcal{I}_\pi(Q) + \log 2}{\log(1/\rho_{\Delta,Q}(\widehat{t}))},$$

with the positive part inserted to make the bound nonnegative. Multiplying by Δ and optimizing over Δ, Q proves (1.13). □

Proof of Theorem 1.5. For the Gaussian location experiment and the reference law $Q_{\tau,m}$,

$$\mathcal{I}_\pi(Q_{\tau,m}) = \int_T D_{\mathrm{KL}}\big(N(v_t, \tau^2 I) \| N(m, \tau^2 I)\big)\, \pi(dt) = \frac{1}{2\tau^2} \int_T \|v_t - m\|^2 \pi(dt) = \mathcal{I}_{\pi,m}.$$

Applying Proposition 1.4 to $\widehat{\Theta}$ and $Q_{\tau,m}$ gives (1.16) under condition (1.15).

It remains to relate this Bayes location error to a Gaussian range. By the comparison-decoder condition (1.14),

$$\|Y - v_{\widehat{\Theta}}\|^2 - \|Y - v_\Theta\|^2 \leq \Omega(\Theta) - \Omega(\widehat{\Theta}) \leq 0.$$

Since $Y = v_\Theta + \tau Z$, expanding and cancelling $\tau^2 \|Z\|^2$ gives

$$\|v_\Theta - v_{\widehat{\Theta}}\|^2 + 2\tau \langle Z, v_\Theta - v_{\widehat{\Theta}} \rangle \leq 0,$$

and hence

$$\|v_\Theta - v_{\widehat{\Theta}}\|^2 \leq 2\tau \langle Z, v_{\widehat{\Theta}} - v_\Theta \rangle = 2\tau\big(X_{\widehat{\Theta}} - X_\Theta\big).$$

Taking expectations and using Jensen's inequality,

$$c^2 \Delta^2 \leq \Big(\mathbb{E} d(\Theta, \widehat{\Theta})\Big)^2 \leq \mathbb{E} d(\Theta, \widehat{\Theta})^2 \leq 2\tau\, \mathbb{E}\big[X_{\widehat{\Theta}} - X_\Theta\big].$$

This proves (1.17). □

Proof of Corollary 1.6. By the comparison condition (1.18) and the identity $Y = v_\Theta + \tau Z$,

$$\|v_\Theta + \tau Z - v_{\widehat{\Theta}}\|^2 \le \|\tau Z\|^2.$$

Expanding the left-hand side and cancelling the common term $\tau^2\|Z\|^2$, we obtain the pathwise inequality

$$d(\Theta, \widehat{\Theta})^2 = \|v_\Theta - v_{\widehat{\Theta}}\|^2 \le 2\tau\langle Z, v_{\widehat{\Theta}} - v_\Theta\rangle = 2\tau\big(X_{\widehat{\Theta}} - X_\Theta\big).$$

On the event in (1.19), which holds with probability at least $1-\delta$ and is uniform over all $t \in T$,

$$X_{\widehat{\Theta}} - X_\Theta \le |X_{\widehat{\Theta}}| + |X_\Theta| \le A(\widehat{\Theta}) + A(\Theta).$$

Hence, on the same event,

$$d(\Theta, \widehat{\Theta})^2 \le 2\tau\big(A(\Theta) + A(\widehat{\Theta})\big),$$

which proves (1.20).

For the expectation bound, let E denote the event $\{\forall t \in T : \ |X_t| \le A(t)\}$. On E, the preceding display gives

$$d(\Theta, \widehat{\Theta}) \le \sqrt{2\tau\big(A(\Theta) + A(\widehat{\Theta})\big)}.$$

On E^c, we use the trivial bound

$$d(\Theta, \widehat{\Theta}) \le D_T.$$

Therefore

$$\begin{aligned}\mathbb{E}d(\Theta, \widehat{\Theta}) &\le \mathbb{E}\left[\sqrt{2\tau\big(A(\Theta) + A(\widehat{\Theta})\big)}\,\mathbf{1}_E\right] + D_T\,\mathbb{P}(E^c)\\ &\le \mathbb{E}\sqrt{2\tau\big(A(\Theta) + A(\widehat{\Theta})\big)} + \delta D_T,\end{aligned}$$

which proves (1.21). The final claim follows by applying the absolute pointwise Gaussian envelope (1.5) to the anchored process $(X_t)_{t \in T \cup \{t_0\}}$ and then restricting back to T. □

3.2 An algorithmic relaxation route to Sudakov-type comparison

By replacing the exact ghost mass with a class-wide small-ball mass and optimizing this small-ball term over hard priors in Theorem 1.5, we obtain a nearest-neighbor route to a Sudakov-type comparison for priors localized at the tested radius.

Corollary 3.1 (Sudakov-type comparison through supremum relaxation)**.** *Under the setup of Theorem 1.5 with the ordinary nearest-neighbor decoder $\Omega \equiv 0$ and the centered reference choice $m = \bar{v}_\pi$, set*

$$q_\pi(\Delta) := \sup_{a \in T} \pi(B_d(a, \Delta)), \qquad L_\pi(\Delta) := \log \frac{1}{q_\pi(\Delta)}.$$

Since $\rho_{\Delta, Q_{\tau, \bar{v}_\pi}}(\widehat{\Theta}_{\mathrm{nn}}) \le q_\pi(\Delta)$, if $L_\pi(\Delta)$ is larger than a universal constant, one may choose $\tau^2 \asymp \mathsf{V}_\pi / L_\pi(\Delta)$ so that (1.15) holds with an absolute $c \in (0,1)$. Relaxing the nearest-neighbor lower bound (1.17) to supremum over the Gaussian process yields the variance-normalized single-radius obstruction

$$\mathbb{E} \sup_{s,u \in T} (X_s - X_u) \gtrsim \Delta^2 \sqrt{\frac{L_\pi(\Delta)}{\mathsf{V}_\pi}}. \tag{3.1}$$

In particular, if the hard prior is localized at scale Δ, meaning $\mathsf{V}_\pi \le C_0 \Delta^2$, then

$$\mathbb{E} \sup_{s,u \in T} (X_s - X_u) \gtrsim_{C_0} \Delta \sqrt{L_\pi(\Delta)}. \tag{3.2}$$

Equivalently, with

$$N^{\mathrm{loc}}_{\mathrm{spread}}(T,d,\Delta;C_0) := \sup_{\pi:\,\mathsf{V}_\pi\le C_0\Delta^2} \frac{1}{\sup_{a\in T}\pi(B_d(a,\Delta))}, \tag{3.3}$$

one obtains, whenever the logarithm is above the universal threshold required by the information condition,

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim_{C_0} \Delta\sqrt{\log N^{\mathrm{loc}}_{\mathrm{spread}}(T,d,\Delta;C_0)}. \tag{3.4}$$

Proof of Corollary 3.1. For the variance-normalized small-ball statement, use the centered reference choice $m = \bar v_\pi$, for which $\mathcal{I}_{\pi,m} = \mathsf{V}_\pi/(4\tau^2)$, and the bound $\rho_{\Delta,Q_{\tau,\bar v_\pi}}(\widehat\Theta_{\mathrm{nn}}) \le q_\pi(\Delta)$. If $L_\pi(\Delta)$ is above a universal constant, choosing $\tau^2 \asymp \mathsf{V}_\pi/L_\pi(\Delta)$ with a sufficiently large absolute constant makes (1.15) hold for some absolute $c\in(0,1)$. Since $X_{\widehat\Theta_{\mathrm{nn}}} - X_\Theta \le \sup_{s,u\in T}(X_s - X_u)$ pointwise in Z, (1.17) gives

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim \Delta^2\sqrt{\frac{L_\pi(\Delta)}{\mathsf{V}_\pi}}.$$

If $\mathsf{V}_\pi \le C_0\Delta^2$, this becomes $\Delta\sqrt{L_\pi(\Delta)}$, with a constant depending only on C_0. Optimizing over such localized priors gives (3.4). □

The preceding corollary directly recovers a Sudakov-scale lower bound under a natural localized hard-prior condition. This is the precise mild condition needed by the nearest-neighbor route: the prior used to certify small balls must also have second moment comparable to the tested radius.

Corollary 3.2 (Localized packing form)**.** *Let $S\subset T$ be 2Δ-separated and assume that, for some $t_0\in T$ and $C_0<\infty$,*

$$S\subset B_d(t_0, C_0\Delta).$$

If $|S| = M \ge M_0$, where M_0 is a universal constant, then

$$\mathbb{E}\sup_{s,u\in T}(X_s - X_u) \gtrsim_{C_0} \Delta\sqrt{\log M}.$$

Proof of Corollary 3.2. Apply Theorem 1.5 with π equal to the uniform distribution on S. Since S is 2Δ-separated, every Δ-ball contains at most one point of S, so

$$q_\pi(\Delta) = \sup_{a\in T}\pi(B_d(a,\Delta)) \le M^{-1}, \qquad L_\pi(\Delta)\ge \log M.$$

The localization assumption gives

$$\mathsf{V}_\pi = \iint d(s,t)^2\,\pi(ds)\pi(dt) \le 4C_0^2\Delta^2.$$

Equation (3.2) yields the claim. □

For comparison with classical covering notation, we record the finite-set fractional-covering duality used in Chen et al. (2024). Recall that the fractional covering number is equivalent to the canonical covering number, up to absolute-constant changes in the covering radius.

Lemma 3.3 (Fractional-covering duality)**.** *For a finite metric space (T,d) and any $\Delta>0$,*

$$N_{\mathrm{frac}}(T,d,\Delta) := \inf_{\mu\in\Delta(T)}\sup_{x\in T}\frac{1}{\mu(B_d(x,\Delta))} = \sup_{\pi\in\Delta(T)}\frac{1}{\sup_{a\in T}\pi(B_d(a,\Delta))}. \tag{3.5}$$

Proof of Lemma 3.3. Let

$$r_* := \sup_{\mu\in\Delta(T)} \inf_{x\in T} \mu(B_d(x,\Delta)).$$

Then

$$N_{\mathrm{frac}}(T,d,\Delta) = \frac{1}{r_*}.$$

Since T is finite, von Neumann's minimax theorem gives

$$\begin{aligned} r_* &= \sup_{\mu\in\Delta(T)} \inf_{x\in T} \sum_{a\in T} \mu(a)\mathbf{1}\{a\in B_d(x,\Delta)\} \\ &= \inf_{\pi\in\Delta(T)} \sup_{\mu\in\Delta(T)} \sum_{x,a\in T} \pi(x)\mu(a)\mathbf{1}\{a\in B_d(x,\Delta)\} \\ &= \inf_{\pi\in\Delta(T)} \sup_{a\in T} \pi(B_d(a,\Delta)). \end{aligned}$$

Taking reciprocals yields (3.5). □

The equality (3.5) is an entropy duality. It optimizes the small-ball term, but it does not remove the factor V_π in (3.1), which comes from the information radius of the Gaussian location channel. There is therefore no contradiction between the fractional-covering lower bounds of Chen et al. (2024) and the localization requirement above: the fractional-covering duality identifies the optimal hard prior for the small-ball obstruction, while the nearest-neighbor Gaussian comparison also requires that this hard prior have controlled second moment at scale Δ. Thus the comparison directly gives the localized obstruction (3.4) and the localized packing form in Corollary 3.2.

The unrestricted classical Sudakov minoration is a separate Gaussian comparison theorem, not a consequence of linear-programming duality alone. Standard proofs use nontrivial Gaussian comparison and convex-geometric duality arguments; see, for example, the expositions of Pollard (2001) and Ledoux and Talagrand (1991, Chapter 3). The full lower half of the Fernique–Talagrand majorizing-measure theorem is stronger and multiscale (Talagrand, 2005). In the present paper we therefore keep the roles separate: the Bayesian algorithmic envelope supplies a one-scale, decision-theoretic, and prior-agnostic primitive, while the unrestricted Sudakov and generic-chaining lower bounds require the classical Gaussian comparison or multiscale localization machinery.

4 A separating example: overparameterized class and pointwise certification

The exact ghost small-ball mass

$$\rho_{\Delta,Q}(\widehat{t}) = \mathbb{E}_{Y\sim Q}\pi\big(B_d(\widehat{t}(Y),\Delta)\big)$$

can be much smaller than the worst-case relaxation

$$q_\pi(\Delta) := \sup_{a\in T} \pi(B_d(a,\Delta)).$$

The following finite-dimensional heterogeneous-orthogonal-basis example separates four quantities: the usual Fano relaxation for a fixed prior, the Bayesian algorithmic lower envelope for an unrestricted norm-regularized estimator, the pointwise Gaussian upper envelope on the estimator-selected subatlas, and the full-class minimax/global Gaussian scale. The estimator is not artificially restricted to the subatlas. It fits over the full ambient class but carries a canonical *minimum-norm* inductive bias, in the same spirit in which nearest-neighbor and interpolating methods require locality or complexity control to generalize (Cover and Hart, 1967; Belkin et al., 2018). The construction is intentionally elementary, yet it captures the essential heterogeneity. The hubleaf subatlas consists of a heavy hub and nearby orthogonal leaves of norm R, together with a farther orthogonal ambient cloud of norm S. In this sense, it augments classical hard-instance geometries, such as Assouad or Hamming-cube constructions, by introducing exactly two distinct norms and a nonuniform prior that places constant mass on the hub and zero mass on the cloud.

Let $M, N \geq 2$, and let

$$g_0, e_1, \ldots, e_M, f_1, \ldots, f_N$$

be orthonormal vectors in $\mathbb{R}^{M+N+1}$. Fix $R > 0$, set

$$\tau := \frac{R}{\sqrt{\log M}}, \qquad S := \frac{1}{8}\tau\sqrt{\log N} = \frac{R}{8}\sqrt{\frac{\log N}{\log M}},$$

and assume

$$\log N \geq 1024 \log M.$$

Then $S \geq 4R$. Define the hub–leaf subatlas, the ambient cloud, and the full ambient set by

$$H_M := \{Rg_0, Re_1, \ldots, Re_M\}, \qquad C_N := \{Sf_1, \ldots, Sf_N\}, \qquad T_{M,N} := H_M \cup C_N,$$

with Euclidean metric $d(s,t) = \|s - t\|_2$. Notice that all points in H_M have the same norm R, whereas all cloud points have the larger norm S. The Bayesian prior is supported on the decision-relevant lower-norm subatlas H_M:

$$\pi(Rg_0) = \frac{1}{2}, \qquad \pi(Re_i) = \frac{1}{2M}, \quad i = 1, \ldots, M, \qquad \pi(Sf_j) = 0, \quad j = 1, \ldots, N.$$

Take

$$\Delta := \frac{R}{3}.$$

Since $S \geq 4R$, every Δ-ball in $T_{M,N}$ is a singleton. Therefore

$$q_\pi(\Delta) = \sup_{a \in T_{M,N}} \pi(B_d(a, \Delta)) = \frac{1}{2}, \qquad \log \frac{1}{q_\pi(\Delta)} = \log 2.$$

Thus the usual Fano small-ball relaxation sees only constant entropy for this fixed prior, because it is dominated by the heavy hub.

Consider the Gaussian location experiment

$$Y = \Theta + \tau Z, \qquad \Theta \sim \pi, \qquad Z \sim N(0, I_{M+N+1}),$$

and define the unrestricted norm-regularized nearest-neighbor estimator[2]

$$\widehat{\Theta}_\lambda(Y) \in \arg\min_{a \in T_{M,N}} \left\{\|Y - a\|^2 + \lambda\|a\|^2\right\}, \qquad \lambda := 64. \tag{4.1}$$

[2]The constant λ is fixed once and for all. It is not tuned to M, N, R, or τ. We only require it to be a sufficiently large numerical constant so that the norm penalty suppresses the high-norm cloud under the ghost law. For concreteness, throughout the example one may take $\lambda = 64$; no optimization of this value is intended.

This is a minimum-norm interpolation-inspired decoder over the full class $T_{M,N}$. No cutoff radius is built into the algorithm; the norm penalty itself makes the cloud unattractive unless the data strongly support it. Note that the estimator $\widehat{\Theta}_\lambda$ satisfies the condition (1.14) of Theorem 1.5 with

$$\Omega(t) = \|t\|^2. \tag{4.2}$$

Indeed, by definition of $\widehat{\Theta}_\lambda$,

$$\|Y - \widehat{\Theta}_\lambda(Y)\|^2 + \lambda\|\widehat{\Theta}_\lambda(Y)\|^2 \le \|Y - \Theta\|^2 + \lambda\|\Theta\|^2,$$

since $\Theta \in T_{M,N}$. Moreover, $\Theta \in H_M$ almost surely, so $\|\Theta\| = R$. Every possible output of $\widehat{\Theta}_\lambda$ belongs to $T_{M,N} = H_M \cup C_N$, whose points have norm either R or S, with $S \ge 4R$. Hence

$$\|\widehat{\Theta}_\lambda(Y)\| \ge R = \|\Theta\| \qquad \text{almost surely.}$$

Therefore

$$\Omega(\widehat{\Theta}_\lambda(Y)) \ge \Omega(\Theta) \qquad \text{almost surely,}$$

and the condition (1.14) in Theorem 1.5 holds.

We use the ghost/reference law

$$Q_\tau := N(0, \tau^2 I_{M+N+1}).$$

Under Q_τ, the scores

$$2\langle Y, a\rangle - (1+\lambda)\|a\|^2$$

for the $M+1$ points in H_M are exchangeable. Moreover, the probability that the regularized decoder selects a cloud point is negligible. Indeed, if a cloud point Sf_j beats the hub point Rg_0, then

$$2\langle Y, Sf_j\rangle - (1+\lambda)S^2 \ge 2\langle Y, Rg_0\rangle - (1+\lambda)R^2.$$

Since $Y \sim Q_\tau$, the left-minus-right random part is Gaussian with variance at most $4\tau^2(S^2 + R^2)$. Using $S \ge 4R$ and $\lambda = 64$, a Gaussian union bound gives

$$Q_\tau(\widehat{\Theta}_\lambda \in C_N) \le N^{-2}$$

for all sufficiently large N. Conditional on the event $\widehat{\Theta}_\lambda \in H_M$, exchangeability gives

$$Q_\tau(\widehat{\Theta}_\lambda = Rg_0 \mid \widehat{\Theta}_\lambda \in H_M) = \frac{1}{M+1}.$$

Consequently,

$$Q_\tau(\widehat{\Theta}_\lambda = Rg_0) \le \frac{1}{M+1} + N^{-2}.$$

Since every Δ-ball is a singleton and π vanishes on the cloud,

$$\begin{aligned}\rho_{\Delta,Q_\tau}(\widehat{\Theta}_\lambda) &= \mathbb{E}_{Y\sim Q_\tau}\pi\big(B_d(\widehat{\Theta}_\lambda(Y), \Delta)\big) \\ &\le \frac{1}{2}Q_\tau(\widehat{\Theta}_\lambda = Rg_0) + \frac{1}{2M}Q_\tau(\widehat{\Theta}_\lambda \in \{Re_1, \dots, Re_M\}) \le \frac{2}{M}\end{aligned}$$

for all large M, N. Hence

$$\log \frac{1}{\rho_{\Delta,Q_\tau}(\widehat{\Theta}_\lambda)} \ge \log M - O(1),$$

whereas the worst-case relaxation for the same prior gives only $\log 2$.

Optimal algorithmic lower bound on the Bayesian decision problem. The information radius under the reference law Q_τ is

$$\mathsf{I}_{\pi,Q_\tau} = \mathbb{E}_{\Theta\sim\pi} D_{\mathrm{KL}}\big(N(\Theta,\tau^2 I) \,\|\, N(0,\tau^2 I)\big) = \frac{\mathbb{E}_\pi\|\Theta\|^2}{2\tau^2} = \frac{R^2}{2\tau^2} = \frac{1}{2}\log M.$$

Combining this with the ghost-mass bound, for all large M,

$$\mathsf{I}_{\pi,Q_\tau} + \log 2 \le \frac{3}{4}\log\frac{1}{\rho_{\Delta,Q_\tau}(\widehat{\Theta}_\lambda)}.$$

Proposition 1.4 therefore gives

$$\mathbb{E}_{\Theta,Y} d(\Theta,\widehat{\Theta}_\lambda(Y)) \ge c\Delta \asymp R$$

for a universal constant $c > 0$. This is sharp for the fixed Bayesian decision problem. When $\Theta \in H_M$, all true parameters have norm R. The norm-regularized rule selects a cloud point with probability at most N^{-2} by the same score comparison used above, now under $Y = \Theta + \tau Z$. Hence

$$\mathbb{E}_{\Theta,Y} d(\Theta,\widehat{\Theta}_\lambda(Y)) \le \sqrt{2}R + N^{-2}S \lesssim R.$$

Thus the Bayes risk of the actual unrestricted estimator is $\asymp R$.

Usual Fano relaxation is vacuous for the fixed prior. By contrast, replacing the exact ghost mass by the worst-case relaxation $q_\pi(\Delta) = 1/2$ gives

$$\Delta\left[1 - \frac{\mathsf{I}_{\pi,Q_\tau} + \log 2}{\log(1/q_\pi(\Delta))}\right]_+ = \Delta\left[1 - \frac{\mathsf{I}_{\pi,Q_\tau} + \log 2}{\log 2}\right]_+ = 0.$$

Thus, for the same prior, channel, and estimator, the usual Fano relaxation is vacuous, while the exact algorithmic ghost mass gives a sharp Bayes estimation lower bound.

Full-class minimax risk has arbitrarily large gap. The full-class minimax risk is much larger. Let

$$\mathfrak{R}_{\mathrm{mm}}(T_{M,N},\tau) := \inf_{\widehat{\theta}} \sup_{\theta\in T_{M,N}} \mathbb{E}_\theta d(\theta,\widehat{\theta}(Y)),$$

where the infimum is over all measurable estimators with values in the ambient Euclidean space. The trivial estimator $\widehat{\theta} \equiv 0$ gives $\mathfrak{R}_{\mathrm{mm}}(T_{M,N},\tau) \lesssim S$. Conversely, restrict the parameter to the cloud C_N and put the uniform prior on C_N. For any estimator $\widehat{\theta}$, define

$$\widehat{J}(\widehat{\theta}) \in \arg\min_{1\le j\le N} \|\widehat{\theta} - Sf_j\|_2.$$

If $d(Sf_J,\widehat{\theta}) < S/2$, then $\widehat{J}(\widehat{\theta}) = J$, since the balls $B(Sf_j, S/2)$ are disjoint. Hence

$$\mathbb{E}_J d(Sf_J,\widehat{\theta}(Y)) \ge \frac{S}{2}\,\mathbb{P}(\widehat{J} \ne J).$$

By Fano's inequality with reference law $N(0,\tau^2 I_{M+N+1})$,

$$I(J;Y) \le \frac{S^2}{2\tau^2} = \frac{1}{128}\log N.$$

Therefore, for all sufficiently large N,

$$\mathbb{P}(\widehat{J} \neq J) \geq 1 - \frac{I(J;Y) + \log 2}{\log N} \geq \frac{1}{2}.$$

It follows that

$$\mathfrak{R}_{\mathrm{mm}}(T_{M,N}, \tau) \geq cS = cR\sqrt{\frac{\log N}{\log M}}.$$

Thus $\mathfrak{R}_{\mathrm{mm}}(T_{M,N}, \tau) \asymp S$, and when $\log N / \log M \to \infty$, the full-class minimax risk is much larger than the Bayes risk for the specified prior. The minimax scale is governed by a rare ambient cloud that is irrelevant to the Bayesian decision problem.

Optimal algorithmically induced Gaussian certificate. Now define the associated Gaussian process

$$X_t := \langle Z, t \rangle, \qquad t \in T_{M,N},$$

using the same standard Gaussian vector Z as in the location experiment. By the previous reasoning (4.2), the comparison condition (1.14) holds with $\Omega(t) = \lambda \|t\|^2$: the true parameter has norm R, and every output has norm either R or $S \geq R$. Thus Theorem 1.5 holds and it gives the algorithmically induced Gaussian certificate

$$\mathcal{G}_{\mathrm{alg}} := \mathbb{E}_{\Theta,Y}\left[X_{\widehat{\Theta}_\lambda(Y)} - X_\Theta\right] \gtrsim R\sqrt{\log M}.$$

Conversely, on the event $\widehat{\Theta}_\lambda \in H_M$, both Θ and $\widehat{\Theta}_\lambda$ lie in H_M, and

$$\mathbb{E} \sup_{s,u \in H_M} (X_s - X_u) \asymp R\sqrt{\log M}.$$

The event $\widehat{\Theta}_\lambda \in C_N$ has probability at most N^{-2} under the true prior, and its contribution is negligible by Cauchy–Schwarz and the standard Gaussian maximal bound on $T_{M,N}$. Therefore

$$\mathcal{G}_{\mathrm{alg}} \asymp \mathbb{E} \sup_{s,u \in H_M} (X_s - X_u) \asymp R\sqrt{\log M}.$$

This also matches the pointwise Gaussian upper envelope on the selected subatlas. Let

$$\nu_Q^H(a) := Q_\tau\{\widehat{\Theta}_\lambda(Y) = a \mid \widehat{\Theta}_\lambda(Y) \in H_M\}, \qquad a \in H_M,$$

be the ghost selection law conditioned on the lower-norm subatlas. By exchangeability, ν_Q^H is uniform on H_M. Applying Theorem 1.1 to the restricted process on H_M, with ambient prior ν_Q^H, gives, up to universal constants and confidence logarithms,

$$\forall t \in H_M: \qquad |X_t| \lesssim R\sqrt{\log M}.$$

Hence the algorithmic ghost-mass lower bound certifies the sharpness of the pointwise upper envelope on the estimator-selected subatlas.

Global Gaussian supremum has arbitrarily large gap. The full ambient Gaussian field has a larger global scale because of the cloud. Restricting only to C_N,

$$\mathbb{E} \sup_{s,u \in T_{M,N}} (X_s - X_u) \geq \mathbb{E} \max_{1 \leq j \leq N} X_{Sf_j} = S\, \mathbb{E} \max_{1 \leq j \leq N} Z_j \gtrsim S\sqrt{\log N}.$$

By the choice of S,

$$S\sqrt{\log N} = \frac{1}{8}\, R\, \frac{\log N}{\sqrt{\log M}}.$$

Thus the global Gaussian scale exceeds the decision-aligned pointwise scale $R\sqrt{\log M}$ by a factor of order $\log N / \log M$. Taking $N = M^K$, this ratio is of order K, and can be made arbitrarily large.

Summary. The above comparison does not contradict classical nearest-neighbor or minimax theory. If one is free to choose a different hard prior, the usual packing argument can put the prior uniformly on the cloud and recover the larger global scale. The point here is different. For the specified Bayesian decision problem, the relevant geometry is the hub–leaf subatlas H_M. The usual Fano relaxation is dominated by the heavy hub and gives no useful lower bound; the full-class minimax risk and the global Sudakov scale are dominated by an ambient cloud irrelevant to the chosen prior; only the algorithmic ghost-mass lower envelope matches the pointwise upper envelope on the estimator-selected geometry. In this decision-theoretic sense, the Bayesian algorithmic lower bound is the right certificate for pointwise-complexity optimality under the specified prior and the actual minimum-norm interpolation-inspired decoder.

5 Finite-Cutoff Renormalization and Pointwise Dimension

This section makes precise the analogy between finite-cutoff renormalization and nonlinear feature learning in Li and Xu (2026). The analogy does not rest merely on the correspondence between composite layers and RG levels. More essentially, both settings exhibit a common *feature/operator inner-product* structure within each composite layer of a fully connected DNN, or within each exponential-family RG level. This correspondence leads to the same proof mechanism: exact finite-step telescoping, a learned or renormalized secant Gram matrix, and a hierarchical prior over local charts and a global atlas.

This is the mechanism that makes the DNN pointwise theory computable, and it is the same mechanism that can be used in a finite-cutoff nonlinear RG analysis. The language is compatible with rigorous Wilsonian RG frameworks such as Brydges and Slade (2015); Bauerschmidt et al. (2019), but the theorem below is an abstract finite-dimensional statement.

Its practical significance in statistical field theory is that RG is fundamentally about controlling coarse marginals after integrating out fluctuations. For globally non-log-concave and anisotropic interacting fields, direct scalar Lipschitz bounds or global convex-geometric arguments are often too crude. The finite-cutoff viewpoint instead allows one to work on stable local charts where the effective action has a curvature lower bound: directions below the finite resolution are absorbed as approximation error, while global nonconvexity is paid for through an atlas cost. We formalize this mechanism as an abstract deterministic complexity theorem and illustrate it through a chartwise nonconvex phase-atlas example, as discussed at the end of Section 5.2.

Deep-network pointwise theory	Finite-cutoff RG analogue
Layer index ℓ	RG level index j
Weights W_ℓ	Level coupling vector g_j
Feature matrix $F_{\ell-1}(W,X)$	Renormalized operator family $O_j(S_j)$
Feature Gram $F_{\ell-1}F_{\ell-1}^\top$	Secant response Gram Γ_j^{sec}
Exact layer telescoping	Exact replacement over RG levels
Outer Lipschitz $M_{\ell\to L}$	Stability of later RG maps $M_{j\to J}$
Active feature eigenspace	Relevant/marginal active operator subspace
Grassmannian prior	Phase/subspace atlas prior
Feature-rank compression	Contraction or truncation of irrelevant directions
Kernel/NTK baseline	Free or massive GFF baseline

Relation to finite-range decomposition and hierarchical RG. The analogy with rigorous renormalization should be understood at the level of finite-cutoff architecture, not as an identity of objects. In the rigorous RG method of Bauerschmidt et al. (2019), a finite-range decomposition writes the covariance as $C = \sum_j C_j$ and turns Gaussian integration into progressive integration over fluctuation covariances at successive RG levels. This is analogous only to the level decomposition underlying a deep network, not to the pointwise-complexity argument itself. Our finite-step telescoping plays the next, comparison-theoretic role: after the nonlinear RG maps have been defined, it compares two RG trajectories by replacing one RG level at a time and thereby exposes a secant response Gram. It thus provides a new tool for analyzing the cumulative effect of multiple nonlinear composition steps.

The hierarchical Φ^4 model is likewise not literally a covering argument. Its nested blocks, local RG maps, and nonperturbative remainders provide a local-to-global organization of a field theory. Hierarchical covering in pointwise complexity serves the statistical analogue: a prior first chooses a phase, rank, or reference subspace, and then covers locally inside that chart. The common principle is that local finite-cutoff tractability must be made compatible with global uniform control. In rigorous RG the bookkeeping is dynamical, through the flow of couplings and remainders; in pointwise complexity it is geometric and Bayesian, through the local ellipsoidal dimension plus atlas cost.

All statements below are finite-dimensional and finite-cutoff. They are intended as a rigorous structural layer, not as a continuum construction of singular interacting fields.

5.1 Pointwise complexity for nonlinear RG maps

Exact finite-step telescoping for multi-level RG maps. Let $\mathcal{S}_0, \dots, \mathcal{S}_J$ be finite-dimensional normed spaces. For each RG level j, let

$$\mathcal{R}_j^{g_j} : \mathcal{S}_j \to \mathcal{S}_{j+1}, \qquad g_j \in B_2(R_j) \subset \mathbb{R}^{p_j},$$

be a nonlinear coarse-graining map. Starting from a fixed $S_0 \in \mathcal{S}_0$, define the RG trajectory

$$S_{j+1}(g) := \mathcal{R}_j^{g_j}(S_j(g)), \qquad S_J(g) = \mathcal{R}_{J-1}^{g_{J-1}} \circ \cdots \circ \mathcal{R}_0^{g_0}(S_0).$$

This notation covers, for example, the exact Wilsonian identity

$$e^{-S_{j+1}(\Phi)} = \int e^{-S_j(\Phi+\psi_j)}\, d\gamma_j(\psi_j).$$

This is the finite-dimensional analogue of integrating out one fluctuation field at one RG level. Perturbative RG expands this map; here we keep the finite map itself.

For two coupling sequences g, g', define the hybrid trajectory

$$S_j^{(g',g;m)} := \mathcal{R}_{j-1}^{g'_{j-1}} \circ \cdots \circ \mathcal{R}_m^{g'_m} \circ \mathcal{R}_{m-1}^{g_{m-1}} \circ \cdots \circ \mathcal{R}_0^{g_0}(S_0),$$

with the evident convention at the endpoints. Then the following identity is exact.

Lemma 5.1 (Exact RG replacement telescoping)**.** *For every g, g',*

$$S_J(g') - S_J(g) = \sum_{j=0}^{J-1} \Big[\mathcal{R}_{J-1:j+1}^{g'}\big(\mathcal{R}_j^{g'_j}(S_j^{(j)})\big) - \mathcal{R}_{J-1:j+1}^{g'}\big(\mathcal{R}_j^{g_j}(S_j^{(j)})\big) \Big], \tag{5.1}$$

where $S_j^{(j)} := \mathcal{R}_{j-1}^{g_{j-1}} \circ \cdots \circ \mathcal{R}_0^{g_0}(S_0)$ and $\mathcal{R}_{J-1:j+1}^{g'} := \mathcal{R}_{J-1}^{g'_{J-1}} \circ \cdots \circ \mathcal{R}_{j+1}^{g'_{j+1}}$.

Proof. Insert the hybrid trajectories that use g up to scale $j-1$ and g' from scale j onward. The difference between two consecutive hybrids changes only the scale-j map. Summing these consecutive differences gives (5.1). ☐

Let ρ be a semimetric on $\mathcal{S}_J$. The next result is the RG analogue of the non-perturbative DNN feature expansion in Li and Xu (2026): the finite replacement identity induces an ellipsoidal metric whose matrices are finite-scale secant response Grams.

Theorem 5.2 (Secant-Gram metric domination for nonlinear RG)**.** *Fix g and a finite neighborhood $\mathcal{N}(g,\varepsilon)$. Assume that for each j:*

(i) the later RG flow is locally stable: for all states U, V arising in $\mathcal{N}(g,\varepsilon)$,

$$\rho\big(\mathcal{R}_{J-1:j+1}^{g'}(U), \mathcal{R}_{J-1:j+1}^{g'}(V)\big) \le M_{j\to J}(g,\varepsilon)\|U-V\|_{j+1};$$

(ii) the scale-j finite replacement is dominated by a positive semidefinite secant Gram $\Gamma_j(g,\varepsilon)$:

$$\big\|\mathcal{R}_j^{g'_j}(S_j^{(j)}) - \mathcal{R}_j^{g_j}(S_j^{(j)})\big\|_{j+1}^2 \le (g'_j - g_j)^\top \Gamma_j(g,\varepsilon)(g'_j - g_j).$$

Then all $g' \in \mathcal{N}(g,\varepsilon)$ satisfy

$$\rho(S_J(g'), S_J(g))^2 \le \sum_{j=0}^{J-1} (g'_j - g_j)^\top G_j(g,\varepsilon)(g'_j - g_j), \qquad G_j := J\, M_{j\to J}(g,\varepsilon)^2 \Gamma_j(g,\varepsilon). \tag{5.2}$$

Proof. Apply Lemma 5.1, the triangle inequality for ρ, the local stability assumption, and the one-step secant-Gram bound. Cauchy–Schwarz gives a factor J when the J increments are summed; this factor is included in the definition of G_j. ☐

How the secant Gram is verified in exponential-family RG steps. Condition (ii) in Theorem 5.2 is a finite-dimensional secant calculation. Fix the state $S_j^{(j)}$ and set

$$F_j(g_j) := \mathcal{R}_j^{g_j}(S_j^{(j)}).$$

If F_j is continuously Frechet differentiable on the finite neighborhood under consideration and the next-scale norm is Hilbertian, then for $\Delta_j = g'_j - g_j$ and $g_{j,t} = g_j + t\Delta_j$,

$$F_j(g'_j) - F_j(g_j) = \int_0^1 DF_j(g_{j,t})[\Delta_j]dt,$$

and Jensen's inequality gives

$$\|F_j(g'_j) - F_j(g_j)\|_{j+1}^2 \le \Delta_j^\top \left(\int_0^1 DF_j(g_{j,t})^* DF_j(g_{j,t})dt \right) \Delta_j. \tag{5.3}$$

Thus condition (ii) holds with the pairwise secant Gram in parentheses, or with any positive-semidefinite Loewner envelope that dominates it uniformly over the finite neighborhood. For an exponential-family RG step

$$\mathcal{R}_j^g(S)(\Phi) = -\log \int \exp\Big\{ -S(\Phi+\psi) - \sum_{a=1}^{p_j} g_a O_{j,a}(\Phi+\psi) \Big\} d\gamma_j(\psi),$$

one has

$$\partial_{g_a} \mathcal{R}_j^g(S)(\Phi) = \mathbb{E}_{j,\Phi,g}[O_{j,a}(\Phi+\psi)].$$

Consequently the secant Gram is the Gram matrix of the finite-scale response functions

$$\Gamma_{j,ab}^{\mathrm{sec}}(g,g') = \int_0^1 \langle m_{j,a}^t, m_{j,b}^t \rangle_{j+1} dt, \qquad m_{j,a}^t(\Phi) := \mathbb{E}_{j,\Phi,g_{j,t}}[O_{j,a}(\Phi+\psi)].$$

This is the precise RG analogue of the learned feature Gram matrix in the DNN pointwise metric.

Compressed response Grams. The scalar Lipschitz choice $\Gamma_j = L_j^2 I$ verifies condition (ii), but it gives no pointwise gain. A useful pointwise-complexity theorem requires an anisotropic response estimate. Let $K_{j+1\to J}$ be a positive-semidefinite quadratic form measuring the sensitivity of the later RG flow from scale $j+1$ to the terminal observable. The propagated response Gram is then

$$G_j^{\mathrm{RG}}(g,g') := \int_0^1 DF_j(g_{j,t})^* K_{j+1\to J} DF_j(g_{j,t}) dt. \tag{5.4}$$

Suppose that the coupling space decomposes as $\mathbb{R}^{p_j} = U_j \oplus U_j^\perp$. Let P_j be an orthonormal basis matrix for U_j. Assume that, uniformly over the finite neighborhood,

$$\|DF_j(\theta)P_j a\|_{K_{j+1\to J}}^2 \le a^\top A_j a \quad (a \in \mathbb{R}^{\dim U_j}), \qquad \|DF_j(\theta)v\|_{K_{j+1\to J}}^2 \le \tau_j \|v\|^2 \quad (v \in U_j^\perp).$$

Then, for every g' in the neighborhood,

$$G_j^{\mathrm{RG}}(g,g') \preceq 2P_j A_j P_j^\top + 2\tau_j I. \tag{5.5}$$

Indeed, write any direction $h \in \mathbb{R}^{p_j}$ as $h = P_j a + v$, with $v \in U_j^\perp$, and use $\|\xi+\zeta\|^2 \le 2\|\xi\|^2 + 2\|\zeta\|^2$. Integrating along the secant path gives (5.5) in the Loewner order. After diagonalizing A_j, the active eigenvalues give the local ellipsoidal effective dimension, while directions with $\tau_j R_j^2 \lesssim u_j^2$ are below finite resolution and do not contribute. In perturbative RG language, U_j is the finite-resolution span of relevant and marginal operator directions, while irrelevant operators contribute only the residual term. In a nonconvex model this compression is imposed chartwise, and the phase/subspace prior below pays the cost of selecting the chart and active subspace.

Hierarchical phase/subspace prior. For a positive semidefinite matrix $G \in \mathbb{R}^{p\times p}$, a Euclidean radius R, and a resolution $u > 0$, set

$$r_{\mathrm{eff}}(G,R,u) := \max\{k : \lambda_k(G)R^2 \geq u^2\}, \qquad V_{\mathrm{eff}}(G,R,u) := \operatorname{span}\{\text{top } r_{\mathrm{eff}} \text{ eigenvectors}\},$$

and

$$d_{\mathrm{ell}}(G,R,u) := \frac{1}{2}\sum_{k=1}^{r_{\mathrm{eff}}(G,R,u)} \log\left(\frac{CR^2\lambda_k(G)}{u^2}\right). \tag{5.6}$$

The following elementary estimate is the local-chart calculation used throughout. For subspaces $V, \bar{V} \subset \mathbb{R}^p$, write

$$\rho_{\mathrm{proj},G}(V,\bar{V}) := \|G^{1/2}(P_V - P_{\bar{V}})\|_{\mathrm{op}}.$$

Lemma 5.3 (Ellipsoidal local chart)**.** *Let $r = r_{\mathrm{eff}}(G,R,u)$, let $V = V_{\mathrm{eff}}(G,R,u)$, and let $\bar{V} \in \mathrm{Gr}(p,r)$ satisfy*

$$\rho_{\mathrm{proj},G}(V,\bar{V}) \leq \frac{u}{4R}.$$

Let $\pi_{\bar{V}}$ be the uniform law on $B_2(2R) \cap \bar{V}$. Then, after changing the universal numerical constant in the radius, uniformly over $g \in B_2(R)$,

$$\log \frac{1}{\pi_{\bar{V}}\{g' : (g'-g)^\top G(g'-g) \leq Cu^2\}} \leq C + d_{\mathrm{ell}}(G,R,u).$$

Equivalently, replacing u by a constant multiple gives the displayed local-chart bound used in Theorem 5.4.

Proof. The point is that the prior may be supported on the reference subspace $\bar{V}$ even though the center g need not belong to $\bar{V}$. Since $r = r_{\mathrm{eff}}(G,R,u)$, the component of any vector in the inactive subspace $V^\perp$ has G-length at most u on the Euclidean ball $B_2(R)$. The projection-distance assumption transfers the active subspace V to $\bar{V}$ with additional G-error at most $u/4$. Hence the G-ball centered at g, with radius enlarged by a universal constant, contains a translate inside $\bar{V}$ of an r-dimensional ellipsoid whose principal semi-axes are comparable to

$$\min\left\{R, \frac{u}{\sqrt{\lambda_k(G)}}\right\}, \qquad k = 1,\dots,r.$$

Because the support of $\pi_{\bar{V}}$ is $B_2(2R)\cap\bar{V}$, this translated ellipsoid lies in the support up to another universal loss. The volume ratio between $B_2(2R)\cap\bar{V}$ and this ellipsoid is therefore bounded by

$$C\prod_{k=1}^{r}\left(\frac{CR\sqrt{\lambda_k(G)}}{u}\right),$$

with factors below one omitted precisely by the definition of r_{eff}. Taking logarithms gives $C + d_{\mathrm{ell}}(G,R,u)$. This is the same finite-dimensional volume-ratio argument used in the DNN local-chart lemma. $\square$

Let $\mathcal{A}_j$ be a finite set of phase labels at scale j, with prior weights $q_j(a) > 0$. Conditional on a phase label a and rank r, let $\nu_{j,a,r}$ be a prior on the Grassmannian $\mathrm{Gr}(p_j,r)$. For $G_j = G_j(g,\varepsilon)$, define the atlas cost

$$\mathfrak{A}_j(g,\varepsilon,u_j) := \log\frac{1}{q_j(a_j(g))} + \log(1+p_j) + \log\frac{1}{\nu_{j,a_j,r_j}\big(B_{\rho_{\mathrm{proj},G_j}}(V_j, u_j/(4R_j))\big)}, \tag{5.7}$$

where $r_j = r_{\text{eff}}(G_j, R_j, u_j)$, $V_j = V_{\text{eff}}(G_j, R_j, u_j)$, and $a_j(g)$ is any chart containing the trajectory at scale j. If $\nu_{j,a,r}$ is Haar-uniform on $\mathrm{Gr}(p_j, r)$, the last term is bounded by the usual Grassmannian entropy; in an ellipsoidal projection metric one may use the anisotropic Grassmannian bound from the DNN pointwise theory (Li and Xu, 2026).

Theorem 5.4 (Finite-scale RG pointwise dimension)**.** *Assume the metric domination* (5.2)*. Choose resolutions $u_j > 0$ such that $\sum_{j=0}^{J-1} u_j^2 \le c\varepsilon^2$. Construct a hierarchical prior Π by independently, at each scale j, sampling a phase label a, an effective rank r, a reference subspace $\bar{V} \in \mathrm{Gr}(p_j, r)$, and then sampling g_j uniformly in $B_2(2R_j) \cap \bar{V}$. Then, uniformly for every trajectory g,*

$$\log \frac{1}{\Pi(B_\rho(g,\varepsilon))} \le C \sum_{j=0}^{J-1} \left[d_{\text{ell}}(G_j(g,\varepsilon), R_j, u_j) + \mathfrak{A}_j(g,\varepsilon,u_j) \right]. \tag{5.8}$$

Proof. For each scale choose the phase label $a_j(g)$, the active rank r_j, and a reference subspace $\bar{V}_j$ within projection distance $u_j/(4R_j)$ of V_j. Lemma 5.3 gives the local prior mass of the ellipsoidal u_j-ball inside this chart. Multiplying the scale-wise prior masses and including the phase, rank, and subspace probabilities gives the right-hand side of (5.8). Finally, if each scale perturbation lies in its ellipsoidal u_j-ball, then (5.2) and $\sum_j u_j^2 \le c\varepsilon^2$ imply $g' \in B_\rho(g,\varepsilon)$. Hence the product event is contained in the ρ-ball, and its prior mass lower-bounds $\Pi(B_\rho(g,\varepsilon))$. $\square$

Interpretation. The theorem is the RG counterpart of the DNN Riemannian-dimension calculation. The local term d_{ell} is the effective dimension of the finite-scale response Gram. The atlas term is the price of making the prior independent of the realized RG trajectory. Irrelevant directions do not contribute once their eigenvalues fall below the finite resolution u_j^2/R_j^2; relevant and marginal directions are precisely the active directions that remain in the sum.

Free Gaussian RG as the fixed-kernel baseline Let $E = E_C \oplus E_F$ be finite-dimensional and let a centered Gaussian field have precision matrix

$$Q = \begin{pmatrix} Q_{CC} & Q_{CF} \\ Q_{FC} & Q_{FF} \end{pmatrix}, \qquad Q_{FF} \succ 0.$$

Define the Schur-complement precision

$$Q_C^{\text{RG}} := Q_{CC} - Q_{CF} Q_{FF}^{-1} Q_{FC}.$$

Then marginalizing the fluctuation variables gives $\phi_C \sim N(0, (Q_C^{\text{RG}})^{-1})$, and conditionally

$$\phi_F = -Q_{FF}^{-1} Q_{FC} \phi_C + \zeta, \qquad \zeta \sim N(0, Q_{FF}^{-1})$$

independently of ϕ_C. This is a fixed-kernel RG step: all pointwise complexity is determined by the Schur-complement covariance and the fluctuation covariance. It is therefore analogous to a kernel, GP, or NTK model rather than to nonlinear feature learning.

5.2 A finite nonconvex phase-atlas example

The previous theorem is abstract. We now give a concrete finite-volume chartwise example. The point is not to construct a full continuum double-well field, but to show how a globally non-log-concave finite measure can be controlled after it is decomposed into stable phase charts. Let $E = E_C \oplus E_F$, let $\mathcal{A}$ be a finite set of phases, and let

$$\nu(d\phi_C, d\phi_F) = \sum_{a\in\mathcal{A}} w_a \nu_a(d\phi_C, d\phi_F), \qquad \nu_a \propto e^{-S_a(\phi_C,\phi_F)} \mathbf{1}_{\Omega_{C,a}\times\Omega_{F,a}} d\phi_C d\phi_F,$$

where the sets $\Omega_{C,a}, \Omega_{F,a}$ are closed convex sets with nonempty interior. The mixture may be globally non-log-concave and multimodal. Assume that, on each chart,

$$\nabla^2 S_a(\phi_C, \phi_F) \succeq H_a = \begin{pmatrix} A_a & B_a \\ B_a^\top & D_a \end{pmatrix}, \qquad D_a \succ 0, \qquad H_a^{\mathrm{RG}} := A_a - B_a D_a^{-1} B_a^\top \succ 0.$$

Theorem 5.5 (One-step nonconvex phase-atlas RG stability)**.** *Under the preceding assumptions, the coarse marginal is again a finite phase-atlas mixture*

$$\nu_C(d\phi_C) = \sum_{a \in \mathcal{A}} w_a^+ \nu_{C,a}(d\phi_C), \qquad \nu_{C,a} \propto e^{-S_a^{\mathrm{RG}}(u)} \mathbf{1}_{\Omega_{C,a}} du,$$

where the weights w_a^+ are the corresponding normalized marginal weights and

$$S_a^{\mathrm{RG}}(u) := -\log \int_{\Omega_{F,a}} e^{-S_a(u,v)} dv.$$

Moreover, S_a^{RG} is H_a^{RG}-strongly convex. Hence, for any finite family of coarse linear observables $Y_x(u) = \langle v_x, u \rangle$, the centered chart process $Y_x - \mathbb{E}_{\nu_{C,a}} Y_x$ is sub-Gaussian with canonical metric

$$d_a(x,y)^2 = \langle v_x - v_y, (H_a^{\mathrm{RG}})^{-1}(v_x - v_y) \rangle.$$

Consequently, the standard sub-Gaussian chaining and variance peeling arguments applied chart by chart, with the comparison metric induced by $(H_a^{\mathrm{RG}})^{-1}$, yields a simultaneous envelope for the mixture, with an additional phase-atlas cost $\log(1/\min_a w_a^+)$, or $\log |\mathcal{A}|$ for equal weights.

Proof. The marginal formula is the definition of integration over E_F. The Schur-complement lower bound follows by completing the square:

$$\begin{pmatrix} u \\ v \end{pmatrix}^\top H_a \begin{pmatrix} u \\ v \end{pmatrix} = u^\top H_a^{\mathrm{RG}} u + (v + D_a^{-1} B_a^\top u)^\top D_a (v + D_a^{-1} B_a^\top u).$$

Moreover,

$$H_a - \begin{pmatrix} H_a^{\mathrm{RG}} & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} B_a D_a^{-1} B_a^\top & B_a \\ B_a^\top & D_a \end{pmatrix} \succeq 0.$$

Hence $S_a(u,v) - \frac{1}{2} u^\top H_a^{\mathrm{RG}} u$ is jointly convex on the product chart. Including the indicators of the convex sets $\Omega_{C,a}$ and $\Omega_{F,a}$, the Prékopa–Leindler theorem implies that $S_a^{\mathrm{RG}}(u) - \frac{1}{2} u^\top H_a^{\mathrm{RG}} u$ is convex on $\Omega_{C,a}$. Thus the coarse chart is H_a^{RG}-strongly log-concave. Brascamp–Lieb and Herbst then give the stated sub-Gaussian bound for linear observables. The envelope follows from the usual generic-chaining upper bound for sub-Gaussian increments, followed by the same pointwise peeling used in the Gaussian proof and a union bound over $a \in \mathcal{A}$. $\square$

Corollary 5.6 (Pure-phase charts for a finite-volume double-well Φ^4 model)**.** *Let $G = (V, E)$ be finite and*

$$S(\phi) = \frac{1}{2} \sum_{\{u,v\} \in E} c_{uv} (\phi_u - \phi_v)^2 + \lambda \sum_{u \in V} (\phi_u^2 - a^2)^2.$$

Fix $m > a/\sqrt{3}$ and define phase charts

$$\Omega_+ = \{\phi : \phi_u \geq m \ \forall u\}, \qquad \Omega_- = -\Omega_+.$$

The restrictions of $e^{-S(\phi)}d\phi$ to Ω_+ and Ω_- define two stable phase charts. A finite mixture of these chart measures is generally non-log-concave, but every chart has Hessian bounded below by

$$L_G + 4\lambda(3m^2 - a^2)I.$$

Therefore Theorem 5.5 applies to every finite decomposition of variables $V = C \sqcup F$, and one nonlinear coarse-graining step preserves a two-chart pointwise-complexity envelope for these pure-phase restrictions. To cover the full double-well Gibbs measure one would need additional charts, for instance droplet or interface charts, and corresponding chartwise Hessian bounds.

Proof. On either chart, $|\phi_u| \geq m$, and

$$\frac{d^2}{dt^2}\lambda(t^2 - a^2)^2 = 4\lambda(3t^2 - a^2) \geq 4\lambda(3m^2 - a^2) > 0.$$

Adding the graph Dirichlet Hessian gives the displayed lower bound. The result follows from Theorem 5.5. □

Significance of the phase-atlas example. The finite nonconvex phase-atlas example should be read as a finite-cutoff bridge between convex concentration theory, high-dimensional geometry, and interacting-field intuition. Within a single stable phase chart, the measure is strongly log-concave: a uniform Hessian lower bound gives Brascamp–Lieb covariance domination and, through logarithmic-Sobolev/Herbst arguments, Gaussian-type concentration for linear and Lipschitz observables (Brascamp and Lieb, 1976; Bakry and Emery, 1985). A mixture of several such charts, however, is generally not log-concave and may be genuinely multimodal, so there need not be a single global convex potential or a single global Gaussian comparison. The phase-atlas decomposition restores tractability by separating these effects. Chartwise, the Schur-complement Hessian gives the local Gaussian comparison metric and hence the pointwise ellipsoidal complexity; globally, one pays an explicit atlas cost for selecting the relevant phase and active subspace.

This distinction is also natural from the viewpoint of convex geometry. Ordinary log-concavity gives powerful isoperimetric and tail estimates, and one-dimensional marginals of isotropic log-concave measures have universal sub-exponential tails (Borell, 1975; Lovasz and Vempala, 2007). It does not, however, by itself provide dimension-free Gaussian-type control in all directions. The Kannan–Lovasz–Simonovits conjecture is a central benchmark for dimension-free isoperimetric and Poincare-type control of isotropic log-concave measures (Kannan et al., 1995). The finite-cutoff viewpoint used here is different. Rather than trying to prove such global dimension-free control for an arbitrary log-concave measure, we work on local charts where the cutoff and the quadratic part of the effective action give a uniform curvature lower bound. On each chart this strong log-concavity yields Gaussian-type comparison in the anisotropic Hessian metric. Global non-log-concavity is then handled by the phase atlas, and directions below the finite resolution are paid for as approximation error rather than controlled through a global Cheeger constant.

In this sense, the pointwise-complexity theorem gives an explicit spectral version of a local convex-geometric principle with global, finite-cutoff extension: the effective spectrum of the chartwise Hessian determines the local ellipsoidal dimension, while the phase atlas records the cost of moving between globally non-log-concave charts. This is parallel to the local-chart/global-atlas construction in the DNN pointwise theory (Li and Xu, 2026), but expressed here in the language of finite-cutoff interacting fields.

5.3 Relation to rigorous RG and constructive QFT

The finite-cutoff theorem above controls metric-ball complexity of an RG trajectory or of chartwise coarse observables. It does not by itself prove Wick factorization, Gaussianity of a scaling limit, or a mass gap. This distinction is essential.

Relation to rigorous finite-range RG. Finite-range decomposition and hierarchical RG provide model-specific mechanisms for constructing and controlling the scale-by-scale maps. In the notation of Bauerschmidt et al. (2019), the covariance decomposition and progressive integration create the RG trajectory; the perturbative coordinates and nonperturbative norms then control its flow. Our theorem starts after this structure has been exposed: it asks what pointwise ellipsoidal metric is induced by the finite-scale response and how the resulting active directions should be covered by a prior. Thus rigorous RG supplies the model-specific dynamical estimates, while the present theorem supplies the local-complexity bookkeeping once a response Gram is available.

Relation to Aizenman–Duminil-Copin. In the four-dimensional Ising and lattice $\lambda\Phi_4^4$ setting, Aizenman and Duminil-Copin (2021) prove marginal triviality by controlling deviations from Wick's rule through random currents and multiscale estimates. This is not primarily an amplitude-control problem: in the Ising case the microscopic spins are already uniformly bounded. The central issue is the Gaussianization of macroscopic observables, namely the decay of the non-Gaussian connected-correlation remainder. Thus Theorem 5.4 cannot reprove their theorem by itself. Its role is complementary: once a model-specific random-current or rigorous-RG argument establishes Gaussianization, the pointwise-complexity theorem supplies a local, finite-cutoff, and observable-sensitive envelope whose cost is carried only by active response-Gram directions and by the phase/subspace atlas.

Universal approximation is not a certificate. One might try to approximate the RG remainder by a neural network, using universal approximation theorems (Cybenko, 1989; Hornik et al., 1989). At a fixed finite cutoff this can approximate a continuous finite-dimensional RG map on a compact set. However, universal approximation alone gives neither a scale-uniform error bound nor preservation of locality, reflection positivity, correlation inequalities, or Wick-factorization errors. To obtain an ADC-type theorem through a DNN-parametrized RG ansatz, one would need certified estimates of the form

$$\|\mathcal{R}_j(S_j) - \widehat{\mathcal{R}}_j(S_j)\|_{\mathcal{N}_{j+1}} \le \eta_j, \qquad \sum_j \eta_j < \infty,$$

in a norm $\mathcal{N}_j$ that controls connected correlations, together with a marginal-flow estimate such as

$$\lambda_{j+1} \le \lambda_j - c\lambda_j^2 + C\eta_j$$

and contraction of irrelevant remainders. These are model-specific RG estimates, not consequences of approximation theory or covering numbers alone.

Yang–Mills. For Yang–Mills, the corresponding finite-cutoff program would have to be formulated directly for lattice gauge fields, rather than for scalar fields. The natural observables are gauge-invariant quantities such as plaquette variables, Wilson loops, and character observables. Chatterjee's probabilistic formulation makes this viewpoint explicit: finite lattice gauge theories are probability measures on edge configurations, Wilson loops are the basic gauge-invariant observables, and mass-gap questions can be phrased through exponential decay of Wilson-loop or plaquette correlations (Chatterjee, 2018). An exact finite-step replacement argument should therefore expose a secant response Gram on the space of such gauge-invariant observables, while the atlas should be compatible with gauge orbits and with the active gauge-invariant operator subspaces selected by the finite-cutoff RG flow. A theorem of this type could organize the finite-cutoff complexity layer of a possible RG approach to Yang–Mills. It would still need to be combined with genuinely model-specific ingredients: construction of the continuum Euclidean field, Osterwalder–Schrader positivity and reconstruction, and a uniform exponential-clustering estimate for gauge-invariant correlations, which is the probabilistic manifestation of a mass gap (Jaffe and Witten, 2000; Chatterjee, 2018). Thus pointwise complexity should be viewed as a finite-cutoff organizing tool for the RG part of the problem, not as a standalone program to prove the Yang–Mills mass gap.

Summary. The common mechanism with nonlinear DNNs is exact finite-step telescoping plus a response Gram plus a data-independent atlas. In DNNs this exposes learned feature-Gram compression. In nonlinear RG it exposes active relevant/marginal operator directions and separates them from below-resolution irrelevant directions. The theorem is therefore a rigorous finite-dimensional complexity layer; scaling-limit Gaussianity, nontrivial continuum interaction, and gauge-theoretic mass gaps require additional model-specific estimates.

6 Graph Local-Time Application versus Cover Time

This section specializes the pointwise Gaussian theorem to pinned Gaussian free fields and then, through Ray–Knight, to random-walk local times. The distinction from cover-time theory is clearest for target questions. A scalar cover-time theorem concerns the event that every vertex has been visited. The field-level Ray–Knight estimate can instead be evaluated on a target field before one passes to any full-cover event. Local time is a standard tool towards cutoff-uniform local integrability control in constructive QFT (Dynkin, 1984).

The phase-star examples below make this distinction quantitative. A target subatlas I is assigned a code length through an atlas prior q. The pointwise local-time route pays this target code length, plus any within-chart fluctuation cost; a full-cover route pays the cost of all N phases. Thus the graph examples make explicit the same local-chart/global-atlas principle that underlies the finite-scale ellipsoidal theory in Section 5.

Graph setup, pinned GFF, and Ray–Knight. Let $G = (V, E, c)$ be a finite connected weighted graph with symmetric conductances $c_{xy} = c_{yx} \geq 0$, where $c_{xy} > 0$ exactly when $\{x, y\} \in E$. Write

$$c_x := \sum_{y \in V} c_{xy}.$$

The weighted graph Laplacian acts on functions $f : V \to \mathbb{R}$ by

$$(Lf)(x) = \sum_{y} c_{xy}\big(f(x) - f(y)\big).$$

We consider the continuous-time random walk, started at a distinguished root v_0, that jumps from x to y at rate c_{xy}/c_x. Its normalized local time is

$$L_t(x) = \frac{1}{c_x} \int_0^t \mathbf{1}\{X_s = x\}\, ds, \qquad \tau(t) := \inf\{s \geq 0 : L_s(v_0) > t\}.$$

The pinned Gaussian free field with root v_0 is the centered Gaussian vector $\eta = \{\eta_x\}_{x\in V}$ such that $\eta_{v_0} = 0$ almost surely and whose covariance is the killed Green function

$$\mathbb{E}[\eta_x\eta_y] = \Gamma_{v_0}(x,y) := \mathbb{E}_x[L_{T_{v_0}}(y)],$$

where T_{v_0} is the hitting time of v_0. Equivalently, this covariance is the inverse of the pinned Laplacian on $V \setminus \{v_0\}$. Its canonical metric is

$$d(x,y) = \big(\mathbb{E}|\eta_x - \eta_y|^2\big)^{1/2} = \sqrt{R_{\mathrm{eff}}(x,y)},$$

where R_{eff} is the effective-resistance metric of the conductance network.

We use the following finite-graph form of the second Ray–Knight theorem (Dynkin, 1984). Let η' be an independent copy of the pinned GFF.

Theorem 6.1 (Second Ray–Knight isomorphism on a finite graph)**.** *With the above normalization, under the product law of the walk and the independent fields (η,η'),*

$$\Big(L_{\tau(t)}(x) + \tfrac{1}{2}\eta_x^2\Big)_{x\in V} \stackrel{d}{=} \Big(\tfrac{1}{2}(\eta'_x + \sqrt{2t})^2\Big)_{x\in V}. \tag{6.1}$$

In particular, $L_{\tau(t)}(v_0) = t$ almost surely.

6.1 Target local times versus cover-time reduction

Target local-time envelopes. Let $H \subseteq V$ be a target set. The proof below only requires a deterministic envelope for the pinned GFF on H. Such an envelope may be obtained either by applying Theorem 1.1 to the restricted field on $H \cup \{v_0\}$, or by applying it once to a larger ambient field and then evaluating the resulting bound on H. This distinction is useful in the phase-star examples, where one ambient atlas prior is fixed before the target subatlas is selected.

Corollary 6.2 (Target-set pointwise local-time envelope)**.** *Let $A_H(\cdot;\delta)$ be any deterministic function satisfying*

$$\mathbb{P}\big\{|\eta_x| \le A_H(x;\delta) \text{ for all } x \in H\big\} \ge 1-\delta$$

for the pinned GFF. Then, under the law of the walk, with probability at least $1-\delta$,

$$\forall x \in H: \qquad \big[t - \sqrt{2t}\,A_H(x;\delta/2)\big]_+ \le L_{\tau(t)}(x) \le t + \sqrt{2t}\,A_H(x;\delta/2) + \frac{1}{2}A_H(x;\delta/2)^2. \tag{6.2}$$

Let

$$\tau_H := \inf\{s \ge 0 : L_s(x) > 0 \text{ for every } x \in H\}$$

be the target-cover time of H. The preceding envelope gives the following target-cover and target-blanket criterion.

Corollary 6.3 (Target cover and target blanket criterion)**.** *Assume that for some $t > 0$, $H \subseteq V$, and $\theta \in (0, 1/\sqrt{2})$,*

$$\sup_{x\in H} A_H(x;\delta/2) \le \theta\sqrt{t}.$$

Then, under the law of the walk, with probability at least $1-\delta$,

$$(1-\sqrt{2}\,\theta)t \le L_{\tau(t)}(x) \le \Big(1 + \sqrt{2}\,\theta + \frac{1}{2}\theta^2\Big)t \qquad \forall x \in H.$$

In particular, $\tau_H \le \tau(t)$ on this event, and

$$\max_{u,v\in H} \frac{L_{\tau(t)}(u)}{L_{\tau(t)}(v)} \le \frac{1+\sqrt{2}\,\theta + \theta^2/2}{1-\sqrt{2}\,\theta}.$$

Proof of Corollary 6.2. Set

$$A_x := A_H(x;\delta/2), \qquad U_x := L_{\tau(t)}(x) + \frac{1}{2}\eta_x^2, \qquad V_x := \frac{1}{2}(\eta'_x + \sqrt{2t})^2, \qquad x \in H.$$

By Theorem 6.1, $(U_x)_{x\in H} \overset{d}{=} (V_x)_{x\in H}$. Define

$$D_x := \frac{1}{2}(\sqrt{2t} - A_x)_+^2, \qquad B_x := t + \sqrt{2t}\,A_x + \frac{1}{2}A_x^2.$$

By the envelope assumption applied to the independent copy η', the event $\{|\eta'_x| \le A_x \text{ for all } x \in H\}$ has probability at least $1-\delta/2$. On this event, $D_x \le V_x \le B_x$ for all $x \in H$. Equality in distribution therefore gives

$$\mathbb{P}\{D_x \le U_x \le B_x \text{ for all } x \in H\} \ge 1 - \delta/2.$$

Applying the same envelope to η itself and taking a union bound, with probability at least $1-\delta$, we also have $|\eta_x| \le A_x$ for all $x \in H$. On the intersection of these events,

$$L_{\tau(t)}(x) = U_x - \frac{1}{2}\eta_x^2 \le B_x,$$

and

$$L_{\tau(t)}(x) \ge D_x - \frac{1}{2}A_x^2.$$

Using nonnegativity of local time and the identity

$$\max\left\{0, \frac{1}{2}(\sqrt{2t} - A_x)_+^2 - \frac{1}{2}A_x^2\right\} = \left[t - \sqrt{2t}\,A_x\right]_+$$

gives (6.2). The event in the conclusion depends only on the walk, so the same probability bound holds under the walk law. □

Proof of Corollary 6.3. Under the displayed small-envelope assumption, Corollary 6.2 gives, simultaneously for all $x \in H$,

$$L_{\tau(t)}(x) \ge (1 - \sqrt{2}\,\theta)t > 0$$

and

$$L_{\tau(t)}(x) \le \left(1 + \sqrt{2}\,\theta + \frac{1}{2}\theta^2\right)t.$$

Thus every vertex in H has been visited by time $\tau(t)$, and the displayed ratio bound follows by dividing the upper estimate by the lower estimate. □

Phase-star hierarchies. We now give a finite graph model that separates direct target-local-time control from a reduction through full cover time. If a deterministic target set is fixed in advance and one is allowed to redo the entire Gaussian analysis on that target alone, then classical generic chaining on the restricted field has the correct target scale. The point here is different. We fix one ambient atlas prior before the target is selected, and then evaluate the same pointwise envelope on any target subatlas. The target scale includes the code length of the selected subatlas under this prior. A proof that first establishes full cover loses this target dependence, because it must cover every phase.

Theorem 6.4 (Phase-star target subatlas versus full cover)**.** *Let S_N be the star graph with root v_0, leaves $\{1,\dots,N\}$, and unit conductances. Let $\mathcal{I}$ be a collection of nonempty subsets of $\{1,\dots,N\}$, and let $q \in \Delta(\mathcal{I})$. For $I \in \mathcal{I}$, set*

$$H_I := I, \qquad \tau_I := \inf\{s \geq 0 : L_s(i) > 0 \text{ for every } i \in I\}.$$

Let τ_{cov} be the cover time of the whole star. Define the ambient prior

$$\mu_q := \frac{1}{2}\delta_{v_0} + \frac{1}{4}\mathrm{Unif}\{1,\dots,N\} + \frac{1}{4}\sum_{I\in\mathcal{I}} q(I)\,\mathrm{Unif}(I). \tag{6.3}$$

Let $A_q(i;\delta)$ be the two-sided pointwise Gaussian envelope from (1.5) applied to the full pinned GFF on S_N, with prior μ_q, with $r_0 = s_0 = 1$, and with constants enlarged if necessary. Then the following hold.

(i) *The pinned GFF satisfies $\eta_1,\dots,\eta_N$ independent $N(0,1)$. Moreover, with probability at least $1-\delta$, simultaneously for all $I \in \mathcal{I}$,*

$$\sup_{i\in I} |\eta_i| \leq C\left[\sqrt{\log\frac{4|I|}{q(I)}} + \sqrt{\log\left(\frac{2+\log\log(N+3)}{\delta}\right)}\right]. \tag{6.4}$$

Consequently, the same deterministic right-hand side may be used as $\sup_{i\in I} A_q(i;\delta)$ in the Ray–Knight local-time envelope.

(ii) *For every $\theta \in (0, 1/\sqrt{2})$, there is $C_\theta < \infty$ such that, for each $I \in \mathcal{I}$,*

$$t \geq C_\theta\left[\log\frac{4|I|}{q(I)} + \log\left(\frac{2+\log\log(N+3)}{\delta}\right)\right] \quad\Longrightarrow\quad \mathbb{P}\{\tau_I \leq \tau(t)\} \geq 1-\delta. \tag{6.5}$$

The probability statement is for the selected target I. The prior μ_q and the Gaussian envelope are fixed independently of that selection.

(iii) *The exact inverse-root-local-time laws are*

$$\mathbb{P}\{\tau_I \leq \tau(t)\} = (1-e^{-t})^{|I|}, \qquad \mathbb{P}\{\tau_{\mathrm{cov}} \leq \tau(t)\} = (1-e^{-t})^{N}. \tag{6.6}$$

Thus the $(1-\delta)$-quantile for covering k prescribed leaves is

$$q_k(\delta) := -\log\left(1-(1-\delta)^{1/k}\right) = \log k + O_\delta(1). \tag{6.7}$$

In particular, $q_N(\delta) - q_{|I|}(\delta) = \log(N/|I|) + O_\delta(1)$. Hence, whenever

$$\log\frac{|I|}{q(I)} = o(\log N),$$

the direct target-local-time route has an inverse-root-local-time scale asymptotically smaller than the scale required by any argument that certifies the same target event only through full cover.

Proof. For the star graph, the pinned Laplacian on the leaves is the identity matrix, so the pinned GFF has covariance I_N. For a leaf i, $\sigma(i) = d(i, v_0) = 1$, and the star metric satisfies $d(i, j) = \sqrt{2}$ for distinct leaves. If $i \in I$, then (6.3) gives

$$\mu_q(\{i\}) \geq \frac{q(I)}{4|I|}.$$

Since the metric diameter of the star leaves is bounded by an absolute constant, the localized Fernique–Talagrand functional satisfies

$$\Phi_{\mu_q}(i) \leq C\sqrt{\log \frac{4|I|}{q(I)}} \qquad (i \in I).$$

The uniform component gives $\mu_q(\{i\}) \geq 1/(4N)$ for every leaf. Hence $\Phi_{\mu_q,*} \leq C\sqrt{\log(N+1)}$. With $r_0 = s_0 = 1$, the peeling multiplicity in Theorem 1.1 is bounded by a constant times $2+\log\log(N+3)$. The two-sided pointwise Gaussian theorem gives (6.4). Applying Corollary 6.3, with the usual confidence split, gives (6.5) after increasing C_θ.

It remains to prove (6.6). During visits to the root, the walk waits an exponential time of rate one before departing and then chooses one of the N leaves uniformly. Since $c_{v_0} = N$, accumulated root occupation time at $\tau(t)$ is Nt. Therefore, per unit normalized root local time, departures from the root form a Poisson process of rate N with independent uniform leaf marks. By Poisson thinning, for each leaf i, the number of excursions from the root to i before $\tau(t)$ is Poisson with mean t, and these counts are independent over leaves. A leaf has positive local time by $\tau(t)$ if and only if it has been chosen at least once. This gives the two identities in (6.6). Solving $(1-e^{-t})^k \geq 1-\delta$ gives (6.7), and the asymptotic formula follows from $(1-\delta)^{1/k} = 1 + k^{-1}\log(1-\delta) + o(k^{-1})$. □

The collapsed star has no within-phase geometry. The next theorem adds a local chart to each phase. It is a graph analogue of a phase/subspace atlas: the prior first pays for selecting a target subatlas, and the effective resistance inside each dense chart suppresses the within-chart fluctuation.

Theorem 6.5 (Decorated phase-star subatlas theorem)**.** *Fix $N, K \geq 1$ and $\kappa > 0$ with $\kappa(K+1) \geq 1$. Let $G_{N,K,\kappa}$ be the conductance graph with root v_0, phase centers $c_1, \ldots, c_N$, and local chart vertices $z_{a,1}, \ldots, z_{a,K}$ for each phase a. The root is connected to each c_a by an edge of conductance one. For each a,*

$$C_a := \{c_a, z_{a,1}, \ldots, z_{a,K}\}$$

is a complete graph with edge conductance κ, and there are no other edges. Let $\mathcal{I}$ be a collection of nonempty subsets of $\{1, \ldots, N\}$, and let $q \in \Delta(\mathcal{I})$. For $I \in \mathcal{I}$, define the decorated target

$$H_{I,K} := \bigcup_{a \in I} C_a.$$

Define the ambient prior

$$\mu_{q,K} := \frac{1}{2}\delta_{v_0} + \frac{1}{4}\mathrm{Unif}\left(\bigcup_{a=1}^{N} C_a\right) + \frac{1}{4}\sum_{I \in \mathcal{I}} q(I)\,\mathrm{Unif}\left(\bigcup_{a \in I} C_a\right). \tag{6.8}$$

Then the following hold.

(i) *The effective resistance within a phase is compressed:*

$$R_{\mathrm{eff}}(x,y) \le \frac{2}{\kappa(K+1)}, \qquad x,y \in C_a, \tag{6.9}$$

while

$$R_{\mathrm{eff}}(v_0,c_a) = 1, \qquad R_{\mathrm{eff}}(v_0,x) \le 1 + \frac{2}{\kappa(K+1)}, \quad x \in C_a. \tag{6.10}$$

(ii) *Let $A_{q,K}(x;\delta)$ be the two-sided pointwise Gaussian envelope from (1.5) applied to the full pinned GFF on $G_{N,K,\kappa}$, with prior $\mu_{q,K}$, with $r_0 = s_0 = 1$, and with constants enlarged if necessary. Then, with probability at least $1-\delta$, simultaneously for all $I \in \mathcal{I}$,*

$$\begin{aligned}\sup_{x\in H_{I,K}} |\eta_x| \le C\Bigg[&\sqrt{\log\frac{4|I|}{q(I)}} + \sqrt{\frac{1}{\kappa(K+1)}\log\frac{4|I|(K+1)}{q(I)}} \\ &+ \sqrt{\log\left(\frac{2+\log\log((N+3)(K+3))}{\delta}\right)}\Bigg].\end{aligned} \tag{6.11}$$

Consequently, for every $\theta \in (0, 1/\sqrt{2})$, there is $C_\theta < \infty$ such that, for the selected I,

$$\begin{aligned}t \ge C_\theta\Bigg[&\log\frac{4|I|}{q(I)} + \frac{1}{\kappa(K+1)}\log\frac{4|I|(K+1)}{q(I)} \\ &+ \log\left(\frac{2+\log\log((N+3)(K+3))}{\delta}\right)\Bigg].\end{aligned} \tag{6.12}$$

implies

$$\mathbb{P}\{\tau_{H_{I,K}} \le \tau(t)\} \ge 1-\delta.$$

(iii) *Let τ_{cov} be the cover time of the full decorated graph, and set $C_N^{\mathrm{cen}} := \{c_1,\dots,c_N\}$.*

$$\mathbb{P}\{C_N^{\mathrm{cen}} \subseteq X[0,\tau(t)]\} = (1-e^{-t})^N. \tag{6.13}$$

Since full cover implies that all phase centers have been visited,

$$\mathbb{P}\{\tau_{\mathrm{cov}} \le \tau(t)\} \le (1-e^{-t})^N. \tag{6.14}$$

Thus any argument that proves target cover only by first proving full cover at confidence $1-\delta$ must use inverse-root-local-time at least $q_N(\delta)$ from (6.7). If, for fixed $\delta \in (0,1)$,

$$\log\frac{|I|}{q(I)} + \frac{1}{\kappa(K+1)}\log\frac{|I|(K+1)}{q(I)} + \log\log((N+3)(K+3)) = o(\log N),$$

then the sufficient scale in (6.12) is $o(q_N(\delta))$, while $\mathbb{P}\{\tau_{\mathrm{cov}} \le \tau(t)\} \to 0$ along that scale.

Proof. The resistance estimate inside C_a is the standard effective resistance of a complete graph with $K+1$ vertices and edge conductance κ, namely $2/[\kappa(K+1)]$. Since c_a is the only vertex of C_a connected to the root, attaching the remaining clique to c_a does not change the effective resistance between v_0 and c_a; hence $R_{\mathrm{eff}}(v_0,c_a) = 1$. The triangle inequality for effective resistance gives (6.10).

Put

$$\alpha_{K,\kappa} := \sqrt{\frac{2}{\kappa(K+1)}}.$$

For $x \in C_a$ and $a \in I$, the ball $B_d(x, \alpha_{K,\kappa})$ contains C_a by (6.9). Under $\mu_{q,K}$, this ball has mass at least $q(I)/(4|I|)$. For smaller radii, the singleton mass is at least $q(I)/[4|I|(K+1)]$. Also $\sigma(x) \leq 2$ by (6.10). Therefore, for $x \in C_a$ with $a \in I$,

$$\Phi_{\mu_{q,K}}(x) \leq C\sqrt{\log \frac{4|I|}{q(I)}} + C\alpha_{K,\kappa}\sqrt{\log \frac{4|I|(K+1)}{q(I)}}.$$

The uniform component in $\mu_{q,K}$ gives singleton mass at least $1/[4N(K+1)]$, and hence $\Phi_{\mu_{q,K},*} \leq C\sqrt{\log((N+1)(K+1))}$. With $r_0 = s_0 = 1$, the peeling multiplicity contributes only the displayed $\log\log((N+3)(K+3))$ term. This proves (6.11). The target-cover implication follows from Corollary 6.3 as in the proof of Theorem 6.4.

Finally, each departure from the root chooses one of the phase centers uniformly, and the same Poisson-thinning argument used in Theorem 6.4 gives (6.13). Since full cover requires every phase center to have been visited, (6.14) follows. The final asymptotic statement follows from $q_N(\delta) = \log N + O_\delta(1)$. □

Relation to classical cover-time theory. For a finite connected conductance graph $G = (V, E, c)$, let τ_{cov} be the first time every vertex has been visited, and write

$$t_{\text{cov}}(G) := \max_{v \in V} \mathbb{E}_v[\tau_{\text{cov}}]. \tag{6.15}$$

A blanket time strengthens cover time by requiring that the walk has accumulated roughly stationary proportions of visits at all vertices. One convenient strong version leads to

$$t_{\text{bl}}(G, \delta) := \max_{v \in V} \mathbb{E}_v[\tau_{\text{bl}}(\delta)]. \tag{6.16}$$

Let $H(u, v)$ be the expected hitting time from u to v, and define the commute-time metric

$$\kappa(u, v) := H(u, v) + H(v, u).$$

Let $\mathcal{C} := \sum_{x \in V} c_x$ be the total conductance. The commute-time identity gives

$$\kappa(u, v) = \mathcal{C}\, R_{\text{eff}}(u, v).$$

For an unweighted graph, $\mathcal{C} = 2|E|$. If $\eta = \{\eta_v\}_{v \in V}$ denotes the pinned GFF, then $\mathbb{E}(\eta_u - \eta_v)^2 = R_{\text{eff}}(u, v)$.

Theorem 6.6 (Ding–Lee–Peres (Ding et al., 2012))**.** *For any connected conductance graph $G = (V, E, c)$ and any fixed $0 < \delta < 1$,*

$$t_{\text{cov}}(G) \asymp \gamma_2\big(V, \sqrt{\kappa}\big)^2 = \mathcal{C} \cdot \gamma_2\big(V, \sqrt{R_{\text{eff}}}\big)^2 \asymp_\delta t_{\text{bl}}(G, \delta). \tag{6.17}$$

Equivalently,

$$t_{\text{cov}}(G) \asymp \mathcal{C} \cdot \Big(\mathbb{E} \max_{v \in V} \eta_v\Big)^2. \tag{6.18}$$

Theorem 6.6 is a global theorem: it identifies the scale of the time needed to cover every vertex. The target local-time estimates above answer a different question. They keep a single ambient field-level envelope and evaluate it on a selected subatlas before any full-cover event is invoked. Theorems 6.4 and 6.5 show that this distinction can be strict. On the star, a selected target I has inverse-root-local-time quantile $\log|I| + O_\delta(1)$, while full cover has quantile $\log N + O_\delta(1)$. The pointwise target envelope pays the atlas code length $\log(|I|/q(I))$; in the decorated star it also pays the compressed within-chart term $(\kappa(K+1))^{-1}\log(|I|(K+1)/q(I))$. Any argument that first passes through the full-cover event necessarily pays the coupon-collector cost associated with all N phase centers. If a target set is fixed in advance, and the Gaussian analysis is allowed to be restricted to that target, then classical generic chaining applied to the restricted field can recover the corresponding target scale. The purpose of Theorems 6.4 and 6.5, however, is different: they construct a simultaneous envelope valid uniformly over all target subsets I. The pointwise formulation is what makes this possible. It retains the dependence on the selected subatlas under a single ambient prior, rather than collapsing the analysis at the outset to the full-cover event.

7 Conclusion

The paper has three linked pointwise/algorithmic and finite-cutoff messages. First, for Gaussian processes, the natural refinement of generic chaining is a simultaneous pointwise envelope for the field, not only a scalar estimate on $\mathbb{E}\sup_x X_x$. This envelope is sharp in expectation after optimizing over the prior and recovers the anchored Fernique–Talagrand scale. It should be useful whenever one wants to retain local field information before taking a final supremum.

Second, the Bayesian algorithmic lower envelope gives a single-radius lower-bound counterpart to pointwise complexity. It is an information-theoretic Bayes-risk bound for every estimator $\widehat{t}(Y)$, expressed through the exact ghost small-ball mass $\mathbb{E}_Q\pi(B_d(\widehat{t}(Y),\Delta))$. The comparison-decoder theorem turns this Bayes distance obstruction into a lower bound on a decision-aligned Gaussian range; conversely, a simultaneous pointwise Gaussian envelope yields an algorithmic upper bound on the same decoding risk. Importantly, we construct a heterogeneous-orthogonal-basis "hub–leaves–cloud" example for norm-regularized nearest-neighbor decoder where the fixed-prior Fano relaxation is vacuous, the algorithmic ghost-mass lower envelope and pointwise upper envelope match on the estimator-selected subatlas, while the full-class minimax risk and global Gaussian scale are much larger. Keeping the algorithmic mass is the key to certifying pointwise complexity locally.

Third, finite-cutoff renormalization illustrates the same structural mechanism in a nonlinear setting. Exact telescoping exposes a response Gram, and a hierarchical atlas converts trajectory-dependent active directions into a data-independent prior. The practical reason to keep this theorem finite-cutoff is that it gives a way to control coarse marginals of globally non-log-concave, anisotropic interacting systems through local strong log-concavity, spectral ellipsoidal complexity, and an explicit phase/subspace atlas. More concretely, the local-time application shows how the pointwise upper envelope can be used before reducing the problem to cover times of finite graphs via the Gaussian free field. Ray–Knight transfers pointwise GFF control to target-set local-time control, and the phase-star examples demonstrate that a selected subatlas can have a smaller intrinsic scale than any route through full cover. Across these settings, the contribution is the same finite-scale complexity layer: expose the correct ellipsoidal metric, keep the relevant local prior mass, and pay the atlas cost only when global uniformity requires it.

Taken together, the results suggest a unified viewpoint. The pointwise Gaussian upper theorem shows that a rich field should be controlled through local envelopes before it is collapsed to a global supremum. The Bayesian algorithmic upper and lower bounds show that, for a fixed prior, reference law, and estimator, the relevant information-theoretic complexity may be the estimator-selected subgeometry rather than the whole ambient class. The finite-cutoff RG framework identify the analogous issue for interacting fields: before a covering argument is meaningful, the nonlinear dynamics must first expose the appropriate response Gram and phase/subspace atlas. Thus the same principle appears in three forms: global worst-case objects capture genuine hardness of the full class, while pointwise and algorithmic objects can certify tractability on the local geometry selected by a field, an estimator, or a renormalization trajectory. This is the sense in which the paper connects field-level generic chaining, algorithmic complexity certificates, and finite-cutoff renormalization.

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